

**COMP1831 - Technologies for Anti-Money Laundering and Financial
Crime**

2025-26 - Term 2

Coursework Report



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Table of Contents

Executive Summary	5
1. Background	5
2. Limitations of Existing AML Frameworks	7
3. The AMLaas Solution: Collaborative Intelligence at Scale	9
3.1 Core Technologies	10
4. Data Access Needs.....	11
5. Approach to Data Privacy, Security, and Governance	12
5.1 Technological Privacy Framework.....	12
6. Government Policies and Standards	13
7. High-Level Architecture and Workflow	14
8. Methodology and Tool mapping to AML Lifecycle	15
1) Placement:	16
2) Layering:	16
3) Integration:	17
9. Operational Feasibility and Scale Estimates	17
Conclusion	18
Table of References	19
Declaration of AI Use	22

List of Figures

Figure 1: Relative proportion of SAR submitted via SAR portal. Source: (SARs Annual Report 2025, no date)	6
Figure 2: Traditional AML Framework	8
Figure 3: Distributed Intelligence architecture	9
Figure 4: High-level Architecture and workflow	14
Figure 5: Methodology workflow	15

List of Abbreviations

AMLaaS	Anti-Money Laundering as a Service
CTGAN	Conditional Tabular Generative Adversarial Network
FCA	Financial Conduct Authority
FedAvg	Federated Averaging
GDPR	General Data Protection Regulation
HIPAA	Health Insurance Portability and Accountability Act
IDV	Identity Verification
MFA	Multi-Factor Authentication
NCA	National Crime Agency
NLP	Natural Language Processing
OFSI	Office of Financial Sanctions Implementation
PII	Personally Identifiable Information
PIPL	Personal Information Protection Law
PSP	Payment Service Provider
PSR	Payment Systems Regulator
RBAC	Role-based Access Controls
RNN	Recurrent Neural Network
SAR	Suspicious Activity Report
TF-IDF	Term Frequency-Inverse Document Frequency
UCM	Unified Case Management
UKFIU	UK Financial Intelligence Unit
UNODC	United Nations Office on Drugs and Crime

Executive Summary

Money Laundering is conceptually simple, but its execution as a financial crime is complex in addition to its difficulty in detection. It holds the potential to disrupt and threaten not only the UK's but any nation's security, prosperity and international reputation. Hence-why, this report outlines a proposal for an AMLaaS provider represented as a UK-incorporated FinTech startup, Sentinels AML Services Ltd (hereafter 'Sentinels') seeking authorized license from the FCA and approval from the PSR.

Current approaches include- isolating the service infrastructures, rule-based transaction monitoring systems deployed individually by each regulated firms, which demonstrably fail in detecting the complex cross-institutional network-based patterns which are the typical characteristics of modern financial crime. Sentinels proposes to address this by proposing a shared, privacy-preserved, AI-powered AML intelligence platform to banks, PSPs and IDV service providers across the UK.

The report details the background of the money laundering problem and shortcomings of existing solutions, the proposed solution, data access requirements, high-level architecture and finally elaboration of tools and methodologies mapped to the AML lifecycle.

1. Background

Money Laundering is the process of disguising criminal proceeds to conceal their illegal origin and integrate them into the legitimate financial system. The UNODC estimates that between 2% and 5% of global GDP is laundered annually, equivalent to USD 800 billion to USD 2 trillion (*Pingwire*, no date).

In the UK, the scale of economic crime is particularly acute. The NCA estimates that serious and organized crime costs the UK economy at least £47 billion annually (National Crime Agency, 2025, p. 08). Even though the law enforcement activities have intensified, the volume of laundered money continues to grow. Over £100 billion is estimated to be laundered annually through the UK or through UK corporate structures (*Economic Crime Plan 2: outcomes progress report*, 2025). The sheer volume of data involved is outrageous; in the 2024-2025 reporting period alone, the UKFIU received over 866,000 SARs (SARs Annual Report 2025, 2025). Interestingly, looking deeply into SARs, the Figure 1 shows that Banking and Financial Services represent the largest proportion of SAR reporters.

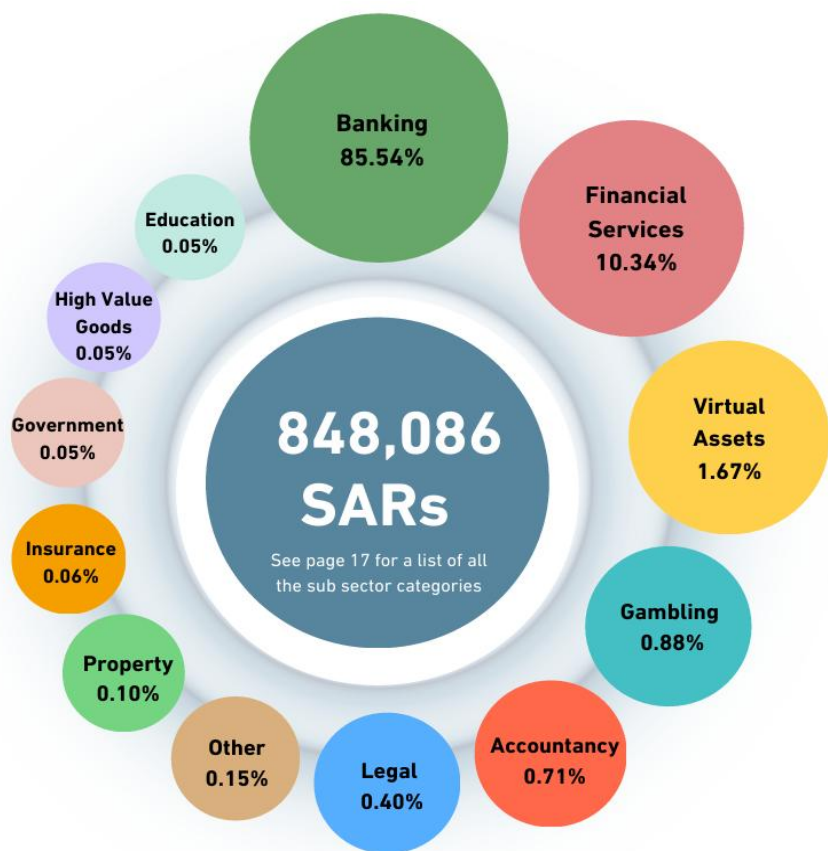


Figure 1: Relative proportion of SAR submitted via SAR portal. Source: (SARs Annual Report 2025, no date)

Some comparative statistics provided by the Table 1:

Table 1: UK Economic Crime and Money Laundering Statistics 2024-2025

Metric	Financial Year 2023-2024	Financial Year 2024-2025	Annual Change	Source
Money Laundering Convictions	3,510	3,756	+7.0%	(Economic Crime Plan 2: outcomes progress report, 2025)
Assets Recovered	£342.7 million	£243.3 million	-29.0%	(Economic Crime Plan 2: outcomes progress report, 2025)

Assets Denied via DAML	£190.3 million	£359.9 million	+89.0%	(SARs Annual Report 2025, 2025)
System-wide Illicit Finance Disruptions	1,990	2,050	+3.0%	(Economic Crime Plan 2: outcomes progress report, 2025)
Firms Compliant with AML Supervision	75%	73%	-2.0%	(Economic Crime Plan 2: outcomes progress report, 2025)

The data suggests a worrying trend i.e., while the number of prosecutions and disruptions is increasing, the effectiveness of the system in terms of asset recovery and overall firm compliance is declining. Similarly, it can be observed that in 2024 the asset recovery value decreases by 29% even though there were higher number of disruptions, which implies that the illicit funds are being transferred with greater speed and concealed using complex layering patterns that the traditional recovery mechanism are failing to address. The traditional rule-based monitoring can no longer catch up to this increase in complexity.

2. Limitations of Existing AML Frameworks

Let's look at the architecture of a typical traditional AML Framework as shown by Figure 2.

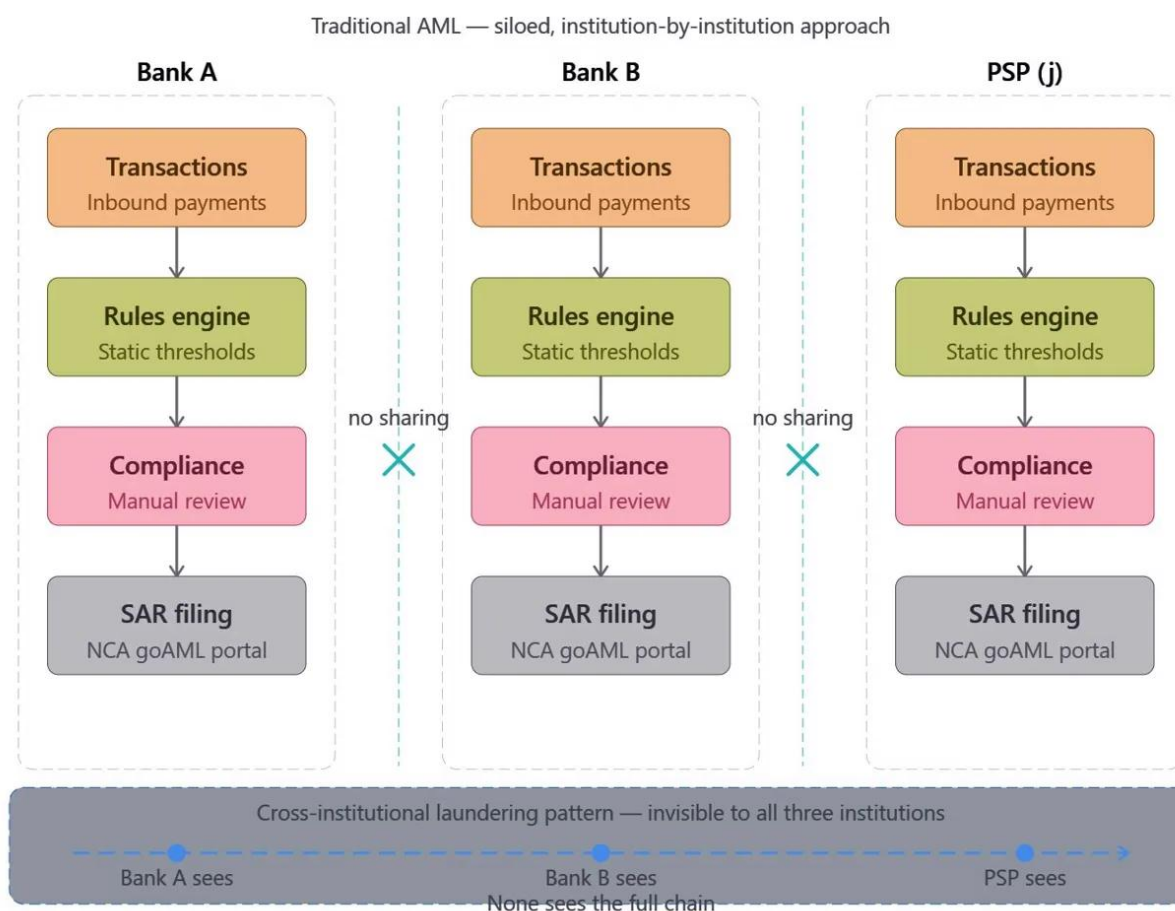


Figure 2: Traditional AML Framework

The fundamental limitation of the current AML solutions is the isolation practice, which in higher level term is referred as the “silo effect”. In the ecosystem consisting k banks, j PSPs, and m IDV service providers, each entity operate with a partial view of any given transaction chain. This provides a fatal gap in the system which is easily exploited by criminals through “smurfing” techniques, breaking down a large illicit lump into hundreds of smaller transactions that fall below reporting thresholds (typically £10,000 in the UK) and distributing them across multiple banks. This is an exploitation of information asymmetries between those entities (Ferwerda and Kleemans, 2019). Even a bank with sophisticated transaction monitoring can only track its own data slice whereas the layering chain between say two banks, a PSP and a foreign correspondent remains invisible to any single entity.

Another issue is, they are heavily rule-based i.e., relying on static thresholds and “if-then” logic. This causes the system to be uselessly reactive and suffer from excessively high false-positive rates

exceeding 90%. These alerts have no operational relevance and only consume analysis time and costs (AML false positives: how technology removes hidden costs, 2026).

Finally, delay in detection is another critical shortcoming. Since they often process data in batch mode, alerts may be generated hours or days after the transaction. But regulators expect real-time monitoring and detection of suspicious activity so they can intercept instantly which is something this rule-based, legacy tools cannot offer (Hosseini, Costa and Carter, 2024).

3. The Solution: Collaborative Intelligence at Scale

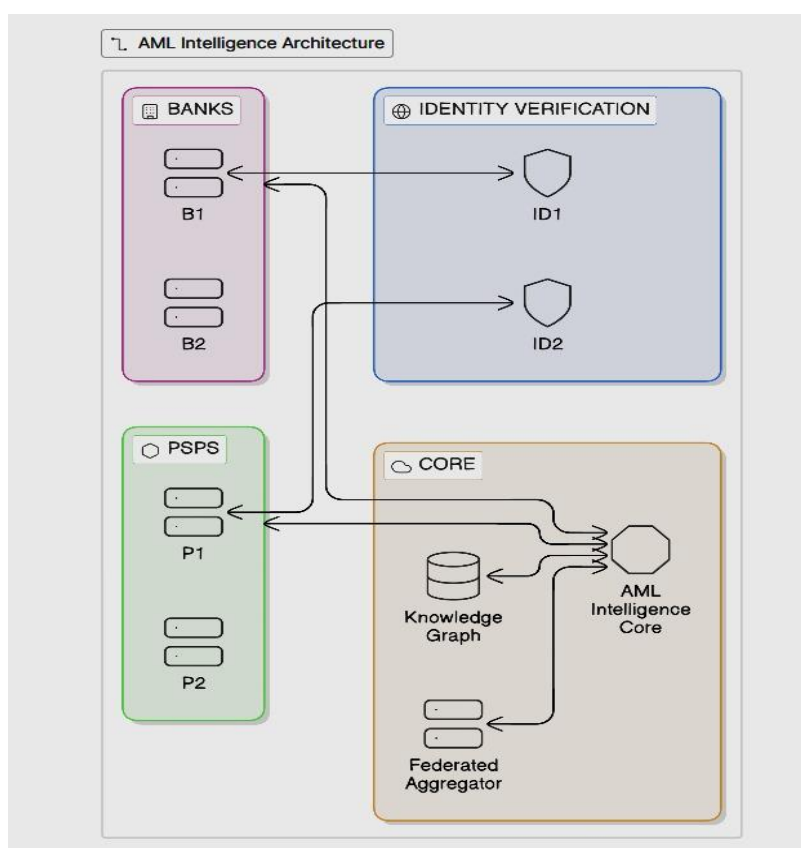


Figure 3: Distributed Intelligence architecture

The gaps reported before is precisely what Sentinels is looking to address, which can be done through a next generation regulatory technology solution. Figure 3 demonstrates what their intelligence architecture will be, i.e., presence of a central intelligence node within the FinTech ecosystem. For the number of k banks and j PSPs in existence, rather than requiring them to build and maintain their own isolated systems, this solution offers AMLaaS through a unified, cloud-native

intelligence layer. In simple words, the regulated firms connect to the platform and contribute pseudonymized transaction data, then the machine learning engine analyzes these signals to identify any suspicious patterns that are not limited by individual institutional boundaries to return risk and other metrics to each participant institutes. This solution acts as an intelligence aggregator not a central massive data lake.

By doing so, this system umbrellas a broad spectrum of user base i.e., from traditional retail banks to robust FinTech and decentralized payment processors, providing them with a real-time, network-wide view of financial crime risk.

3.1 Core Technologies

Since Sentinels' proposition focuses strongly on collaborative monitoring, it is important that the intelligence between institutions be shared without compromising the privacy of raw customers data, hence a technological stack befitting of this is:

- a. **Federated Learning (FL):** A decentralized machine learning paradigm is employed where the data resides on the servers of the participating banks and PSPs. Instead of centralizing the sensitive PII, the AMLaaS provider sends a global model to each participant who then train the model locally on their own unique dataset, and only return the encrypted model updates (gradients) to the central server for aggregation (Bhoyar, 2025). The final updated version of the global model is obtained via the FedAvg algorithm. This “cross-silo” FL approach allows the model to learn from a multi-institution dataset, helping it learn patterns which otherwise would be impossible for a single bank, without needing the raw data ever to leave the institution's perimeter (Banse, Kreischer and Jürgens, 2024).
- b. **Differential Privacy (DP):** The system provides a mathematical guarantee against the leakage of individual data points during model updates by integrating ϵ -differential privacy. By adding Gaussian noise to the local updates before aggregation, it prevents inference of individual training examples from the aggregated model (Bhoyar, 2025). This also allows the platform to achieve high accuracy in detecting suspicious behavior while remaining compliant with the strict regulations of GDPR, HIPAA, PIPL and others (Prakash, 2024; Baligodugula and Amsaad, 2025).
- c. **Graph Databases and Network Analytics:** Even criminals know that breaking large movements of money into many smaller parts and routing these funds through multiple intermediaries (i.e., Structuring and Layering in AML) can slip past monitoring tools since

they appear harmless until all the pieces are connected. Hence, Sentinels plans to utilize high-performance graph databases (e.g., Neo4j or TigerGraph) to model the ecosystem as a web of interconnected entities, transactions, and devices (Shrivastava, 2026). By representing accounts and transactions as nodes and edges, it can perform multi-hop analysis to identify circular patterns like fraud rings and layering chains that are otherwise invisible to relational databases.

- d. **Agentic AI and Visual Analytics:** Sentinels also aims to deploy autonomous agents as “Digital Workers” who not only flag simple anomalies, also gather data, synthesize information, and help drive investigative conclusions which allow investigators and analysts retain oversight and greater reporting power by drafting comprehensive SARs. The aim is not to replace human expertise instead amplify it by freeing teams from manual triage and shift their priorities on complex cases and true risk. Additionally, these are supported by high-fidelity visual analytics that allow analysts to fly through complex transaction networks to identify clusters of high-risk activity.

4. Data Access Needs

The most significant step for the effective operation as an AMLaaS service provider comes down to the approval regarding data access from the PSR and the FCA. The request is not for the centralization of raw customer records, but for the orchestrated access to specific, high-value metadata and cryptographic hashes necessary for network-wide entity resolution and transaction monitoring. Requested data can be categorized as:

- a. **Inter-Bank Transactional Metadata:** Real-time access to the metadata of transactions that occur through the Faster Payments, Bacs, and CHAPS systems is required which includes timestamps, amounts, currency types and various other specifications about the transaction that can provide contexts like payment purpose etc.
- b. **Hashed Identity Identifiers:** Entity Resolution helps to identify data records across multiple sources that refer to the same real-world entity and the link between them. To perform this between banks and PSPs, cryptographic hashes of unique identifiers like passport numbers, NI numbers, device IDs, etc. needs to be accessed. This helps to recognize when a person is using multiple accounts from different institutions to conduct “smurfing” without needing to look at the original ID number.

- c. **Beneficial Ownership and Corporate Linkages:** The “Layering” phase of Money Laundering heavily utilizes shell companies, fund transfers between multiple banks and accounts. Hence, to uncover the true owners of shell companies, transparency is essential. The U.S Corporate Transparency Act in effect has made beneficial-ownership data now part of CDD/KYB workflows (Keene, 2025).
- d. **Global Sanctions and Asset Review:** The platform requires continuous, low latency feeds from the OFSI and international bodies to ensure real-time screening against asset freezes and high-risk individuals (Economic Crime Plan 2: outcomes progress report, 2025).
- e. **Historical SAR Outcomes:** AML systems must evolve with new laundering techniques and regulations. Anonymized NCA feedback on SAR outcomes can train ML models and support reinforcement learning, improving risk detection and reducing false positives.

The data access needs outlined above all align with UK’s National Payment Vision of more streamlined, innovative and secure regulatory environment which can support the digitalization of financial markets.

5. Approach to Data Privacy, Security, and Governance

The security and integrity of sensitive financial information are the key priorities of the proposed service. Sentinels’ architecture follows a privacy-by-design approach, ensuring that even if the central system is compromised, participants’ raw customer data remain protected within their own environments.

5.1 Technological Privacy Framework

To safeguard the FinTech ecosystem, the platform employs a multi-layered security stack such as:

- **End-to-End Encryption and Secure Aggregation:** The banks and central AML service collaborate effectively to train the global model using homomorphic encryption as well as the implementation of a secure multiparty computation protocol based on secret sharing. Since they also implement secure aggregation protocol ensuring that the central server can only see the combined average of model updates from all k participants, making it impossible to reverse engineer the specific data of any individual (Bhoyar, 2025).
- **Zero-Trust Architecture:** The platform assumes that no entity within the network is inherently trusted. The AMLaaS intelligence dashboard’s access will be secured and controlled via MFA

and RBAC in line with the requirements outlined by FCA Senior Management and Systems and Controls (SYSC) (FCA Handbook - SYSC 4.1 General requirements, no date).

- **Differential Privacy Budgeting:** The system monitors an interesting concept called the “privacy budget” (ϵ) for every training round. It is basically an upper bound or limit set to make sure that if a specific query or training session risks exceeding this then the system will automatically halt the process or increase the noise injection (Banse, Kreischer and Jürgens, 2024).

6. Government Policies and Standards

The proposed system will fully comply with the strict governance framework of the FCA handbook, specifically addressing the requirements for third-party outsourcing and financial crime systems such as:

- **FCA SYSC 6.3 Compliance:** System’s policies and procedures will “enable it to identify, assess, monitor and manage money laundering risk” as well as “comprehensive and proportionate to the nature, scale and complexity of its activities” (FCA Handbook - SYSC 6.3 Financial crime, no date).
- **FCA SYSC 8.1 Outsourcing Requirements:** As an outsourced service provider, the startup will comply with this to provide the FCA and its clients with effective access to data with rights to conduct audits and inspections. It will also conduct a thorough risk assessment and report the arrangement to the FCA (Swan Law, 2025).
- **Regulatory Reporting and Audit Trails:** Every automated decision and investigator-led inquiry will generate an immutable audit trail which is the system’s way to answer the regulators demand for a transparent, auditable reasoning model for every flagged transaction or SAR draft (Hosseini, Costa and Carter, 2024).
- **Exit Strategy and Continuity:** In accordance with SYSC 13.9, the platform maintains a contingency arrangements where participating banks can transition their services back in-house or relocate them to another provider ensuring minimal disruption to operations and client services, reducing risks further (FCA Handbook - SYSC 13.9 Outsourcing, no date; Swan Law, 2025).

7. High-Level Architecture and Workflow

Sentinels' system architecture is designed to integrate seamlessly with the UK's payments infrastructure. It operates as a high-performance distributed network using a modular approach with defined systematic nodes.

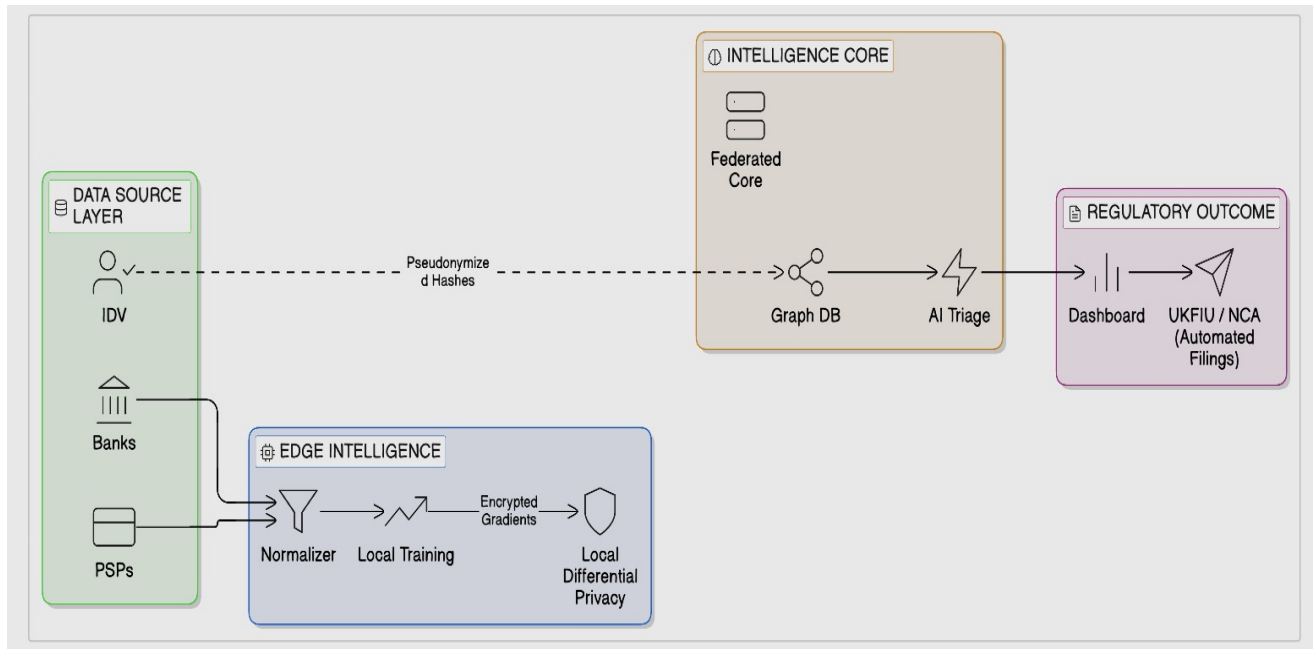


Figure 4: High-level Architecture and workflow

Architectural Breakdown:

- **Data Source Layer:** This is the foundational input tier for the whole ecosystem because they provide the raw input necessary for network-wide intelligence. This layer is composed of some k number of banks, j PSPs and m IDV services. Banks and PSPs are the primary sources of transactional metadata, IDVs contribute with high-value identity signals by supplying Pseudonymized Hashes rather than raw PII. In Sentinels' architecture, this layer stands out as the raw data residency remains within the source institution's perimeter ensuring only normalized metadata and encrypted model gradients are shared to the intelligence core.
- **Edge Intelligence Layer:** This is where each bank and PSPs run a localized container which performs initial data pre-processing tasks, feature engineering and local model training.
- **The Federated Intelligence Core:** The central layer which coordinates the learning process. It distributes the global model to the Edge nodes, collects the differential privacy updates, and aggregates them using the FedAvg algorithm (Banse, Kreischer and Jürgens, 2024).

- **The Central Knowledge Graph:** This layer monitors the whole ecosystem's network topology. It stores hashed entity links as well as cross-bank transaction paths. It also utilizes graph databases like Neo4j and TigerGraph to trace funds across numerous institutions using multi-hop queries.
- **AI-Driven Case Management:** The Agentic AI is adopted as the digital worker analysing network insights, transactional patterns, and smart scoring and screening into a UCM dashboard. This AI-assisted triage drastically reduces false positives by up to 90% and speed up case handling by 40%, allowing compliance teams to focus on high-fidelity alerts rather than rule-based noise (Ibitola, 2025; AML Transaction Monitoring Software Explained: Key Features & Best Practices, 2025).

Workflow Summary:

Basically, the flow starts when a transaction is initiated in a PSP or bank, then the local Edge node instantly scores the transaction based on the local model and historical context. Say the score exceeds a certain threshold then, the Central Knowledge Graph is queried using pseudonymized hashes to see if the transaction pattern is familiar and not a red flag like a part of a layering chain with multiple hops and cross-institutional pattern. If it is confirmed to show a high-risk pattern then an alert is generated in the UCM dashboard, with the necessary investigative summary and details by AI.

8. Methodology and Tool mapping to AML Lifecycle

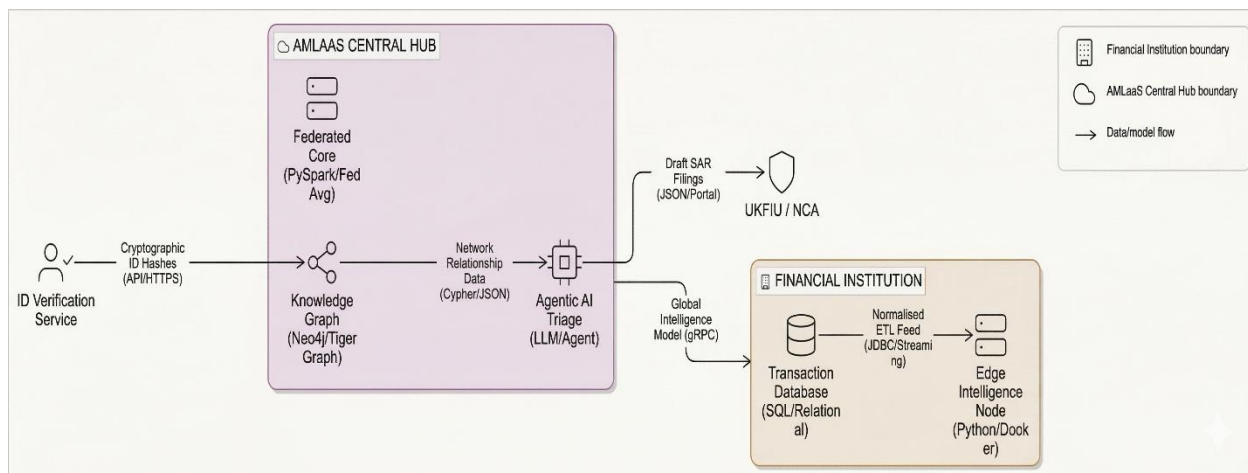


Figure 5: Methodology workflow

As mentioned before, there are three distinct stages in the money laundering lifecycle: Placement, Layering, and Integration.

1) Placement:

It is the process of introducing illicit funds into the financial system. It includes structuring cash deposits below the reporting thresholds by smurfing to avoid triggering suspiciousness (*Pingwire*, no date).

- **Data Preprocessing Methodology:** The obvious problem to combat is the “class-imbalance” where the legitimate transactions easily outnumber fraudulent ones. Sentinels plans to use CTGAN to generate realistic synthetic samples of structuring behavior. With this, the ML model can be trained in balanced dataset, significantly improving the recall rate for low laundering events.
- **Detection Tool:** SQL-based Feature Engineering is used to create dynamic behavioral variables, such as rolling averages for cash deposits over 90 days measuring the frequency of new account openings across different banks.
- **Analytics:** Unsupervised learning method (Clustering) is used to identify outliers that do not match the typical pattern of customers such as a high-volume transaction with no clear economic purpose.

2) Layering:

Layering stage is represents the most complex stage of money laundering, designed to create distance between criminal activity and laundered funds through multiple financial transactions (*Pingwire*, no date).

- **Data Preprocessing Methodology:** System utilizes NLP and TF-IDF to transform unstructured data such as customer service notes or payment references into structured format that the model can leverage and analyze for signs of over-invoicing or fake contracts.
- **Detection Tool:** Graph-based algorithms like Louvain modularity, community detection aids in identifying laundering rings, mule accounts and layered transaction paths (Andreescu *et al.*, 2025). Unlike simple threshold breaches, layering creates a multi-step transaction chain which becomes visible only when analyzed as a network.
- **Analytics:** Multi-Hop Transaction Tracing helps the system trace funds which are split and recombined across multiple institutions. Through this, circular movements where the money

is sent from Bank A to PSP B, through several intermediaries, only for it to be returned to a related entity at Bank C, can be uncovered (Shrivastava, 2026).

3) Integration:

This the final stage of money laundering where successfully laundered money re-enters the legitimate economy as apparently legal assets (*Pingwire*, no date).

- **Data Preprocessing Methodology:** The most critical step in this stage is Entity Resolution. The system uses Fuzzy matching and Label Propagation graph algorithm to link multiple personas used by criminals by analysing their interconnected relationships such as banking details, investments, corporate identities, hidden shell company beneficiaries to form a single 360-degree view of the customer.
- **Detection Tool:** Algorithms like PageRank and Centrality Algorithms helps in detecting liaisons and more relevant players indulging in this act. Accounts that receive fragments of payments through disparate sources and then execute a large transaction or investments are flagged as potential suspects.
- **Analytics:** RNNs have the capability to model dependencies in sequential data and analyze the temporal flow of funds, which helps in distinguishing a legitimate business with predictable cash flows against fraudulent ones where integration transactions patterns occur in a suspicious manner with illogical economic timing.

9. Operational Feasibility and Scale Estimates

Managing the operation costs and complexities of a national-level project is challenging. Although Sentinels is technology-driven, it requires a strong operational structure, sufficient staffing and well-planned infrastructure. Some estimated operational requirements are:

- a) **Headcount and Expertise:** A rough estimation of specialized workforce constituent of 120-150 professionals seems reasonable. The emphasis would undoubtedly be on data science (30% coverage), distributed engineering (40%), and regulatory compliance or investigative officers (20%), with the remainder covering other specialized roles. A key non-negotiable role is human-in-the-loop investigators, where experienced skilled analysts provide feedback to ML models, creating a healthy continuous learning loop from real SAR outcomes and helping reduce false-positives.

- b) **Infrastructure and Cost:** The cloud infrastructure must be able to handle real-time data ingestion and processing which could be in millions every hour so for that purpose, streaming platforms like Apache Kafka seems to be a good option. Similarly, memory-optimized graph database engines like AWS Entity Resolution and Amazon Neptune Analytics can be utilized for executing powerful graph algorithms like Louvain to identify potentially fraudulent activities during the milliseconds of a transaction's authorization window (Hung, Owusu and Gabay, 2026). Although initial investment for this type of platform will be high, the FL model can provide significant long-term savings by distributing the computational burden of model training to different local infrastructures. By doing so, it potentially reduces cross-cloud data transfer cost up to 87% compared to centralized models (Prakash, 2024).

Conclusion

The modern state of Money Laundering speaks for itself. There's a clear demonstration that the traditional, siloed architectures can no longer keep up with the pace of the cyber-enabled frauds, deepfake identity attacks, and complex smurfing and layering techniques that criminals have been using. The integrity and reputation of UK's financial ecosystem is getting tarnished so, a response that is as technologically advanced as the threat it needs to counter is much needed.

Hence, Sentinels' proposed system offers a solution that balances the need for collaborative intelligence with the absolute requirement of data privacy and safety. It can identify patterns of criminal activities that otherwise seem invisible to the traditional systems, at the same time offer to reduce alert fatigue caused due to noise and have enough combination of intelligence in terms of automation but also human intervention where needed.

Authorizing the license will provide the PSR and FCA with a powerful project to achieve their mission of ensuring that UK payment systems remain the safest, most competitive, and most innovative in the world.

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Declaration of AI Use

I have used AI while undertaking the assignment in the following ways:

- To develop research questions on the topic – YES
- To create an outline of the topic – YES
- To explain concepts – YES
- To support my use of language – YES
- In other ways, as described below:
 - To assist in making architectural diagrams and workflow.

Lab Exercises

Lab-1:

Exercise-1.1: Generate some graph data using Cypher.

The cypher query:

```
//Required nodes creation for university, two modules, seven students
and three technologies.

CREATE (uni: University {name: "University of Greenwich" } )
CREATE (module1: Module {name:"COMP1830"}), (module2: Module {name:
"COMP1831" })
CREATE (student1: Student {name:"Rasvan"}), (student2: Student
{name:"Rajesh"}), (student3: Student {name:"George"}), (student4:
Student {name:"Peter"}), (student5: Student {name:"Olivia"}),
(student6: Student {name:"Katie"}), (student7: Student {name:"Julia"})
CREATE (tech1: Technology {name: "Neo4J"}), (tech2: Technology
{name:"Python"}), (tech3: Technology {name:"ML"})

//Linking every student to the University
MATCH (s: Student), (u:University)
CREATE (s)-[:STUDIES_IN]->(u)

//Assigning students to modules(COMP1830 and COMP1831)
MATCH (s: Student), (module: Module {name:"COMP1830"})
WHERE s.name in ["Rasvan","Rajesh","George"]
CREATE (s)-[:ATTENDS]->(module)

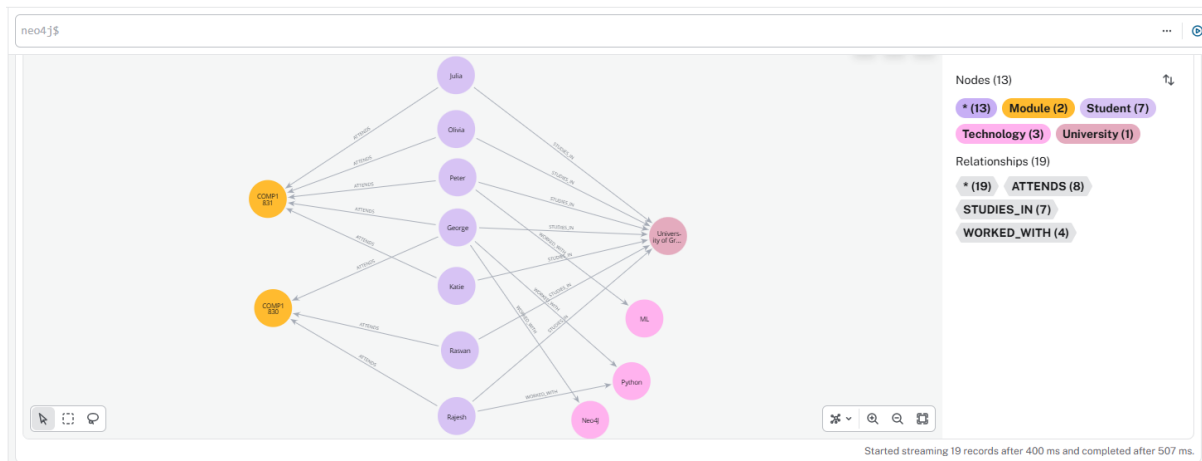
MATCH (s: Student), (module: Module {name:"COMP1831"})
WHERE s.name in ["Katie","Peter","Olivia","George","Julia"]
CREATE (s)-[:ATTENDS]->(module)

//Connecting students that work with (Python, Neo4J and ML)
technologies
MATCH (s: Student {name: "George"}), (tech: Technology)
WHERE tech.name in ["Python", "Neo4J"]
CREATE (s)-[:WORKED_WITH]->(tech)

MATCH (s: Student {name: "Peter"}), (tech: Technology)
WHERE tech.name in ["ML"]
CREATE (s)-[:WORKED_WITH]->(tech)

MATCH (s: Student {name: "Rajesh"}), (tech: Technology)
WHERE tech.name in ["Python"]
CREATE (s)-[:WORKED_WITH]->(tech)
```

Output:



The Cypher query first creates nodes for one university, two modules (COMP1830 and COMP1831), seven students, and three technologies (Neo4J, Python, and ML). It then establishes relationships where all students ‘STUDIES_IN’ the university, certain students ‘ATTENDS’ specific modules, and selected students ‘WORKED_WITH’ certain technologies. This builds a simple graph showing how students are connected to their university, modules, and the technologies they have used as shown above.

Exercise-1.2: Execute a Cypher query

The cypher query:

```
MATCH (s: Student)-[:ATTENDS]->(m: Module {name: "COMP1831"}),
(s)-[:WORKED_WITH]->(t1: Technology {name: "Neo4J"}),
(s)-[:WORKED_WITH]->(t2: Technology {name: "Python"})
RETURN s.name
```

Output:

Table RAW		Q N ~	
s.name			
1	"George"		

This Cypher query matches students who attend the module COMP1831 and have worked with both Neo4J and Python technologies. The query then returns the names of those students who satisfy all these relationships, which was only “George”.

Exercise-2.1: Screenshot of graph structure displaying the imported data

Query to import data:

```
//Creating constraints to have unique Id so that duplicate data won't
be imported
CREATE CONSTRAINT personIdConstraint FOR (person:Person) REQUIRE
person.personId IS UNIQUE
```



```

CREATE CONSTRAINT loanIdConstraint FOR (loan:Loan) REQUIRE loan.loanId
IS UNIQUE

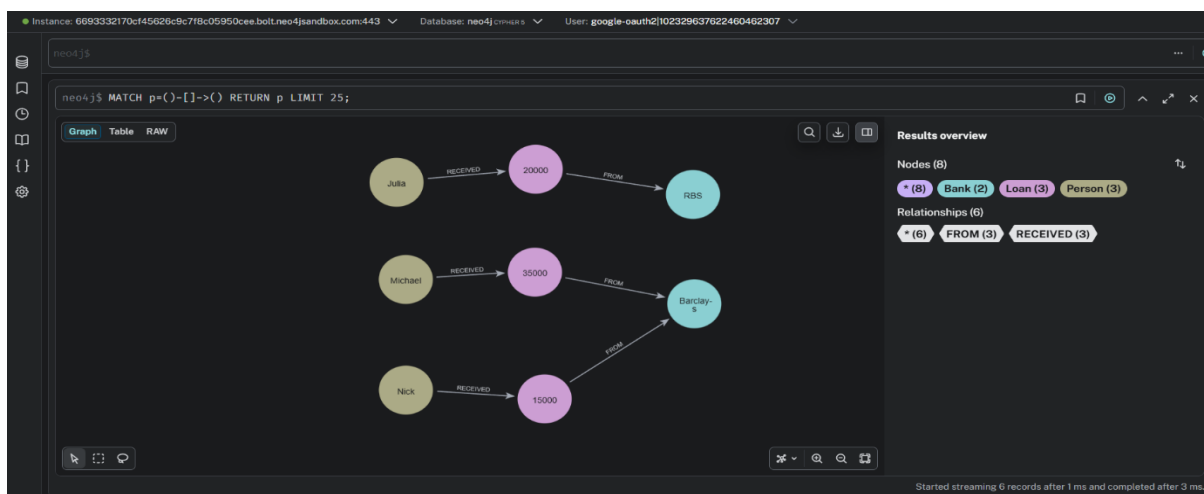
//Importing Persons data
LOAD CSV WITH HEADERS FROM
"https://gist.githubusercontent.com/apogiatzis/237146155b85be73e25d775
e487f4d8c/raw/7a440b1f4b7885ea425d5db0ce6aae041b8ea42a/persons2.csv"
AS csvLine
CREATE (p:Person {id: toInteger(csvLine.personId), fname:
csvLine.firstName, lname: csvLine.lastName})

//Importing loans data
LOAD CSV WITH HEADERS FROM
"https://gist.githubusercontent.com/apogiatzis/93927b266820ce9696b9405
5e896b1f9/raw/fdfdd19194b2afc1f78b01e4ddd783826b676878/loans.csv" AS
csvLine
MERGE (bank:Bank {bname: csvLine.bank})
CREATE (loan:Loan {id: toInteger(csvLine.loanId), amount:
csvLine.amount})
CREATE (loan)-[:FROM]->(bank)

//Importing Loan status data and matching them to persons
LOAD CSV WITH HEADERS FROM
"https://gist.githubusercontent.com/apogiatzis/dbc8f04359311257cd84e21
6ae8fedc4/raw/19fa315206573cab427d8b13e641c403a45852c6/status.csv" AS
csvLine
MATCH (person:Person {id: toInteger(csvLine.personId)}), (loan:Loan
{id:
toInteger(csvLine.loanId)})
CREATE (person)-[:RECEIVED {loan: csvLine.status}]->(loan)

```

Output:



Viewing the graph structure of the imported data showed people with their respective loan status and their loan amounts mapped to which Bank they received their loan from as well.

Lab-2:

Exercise-1.1: Query for identifying financial risk per fraud ring

The Cypher query:

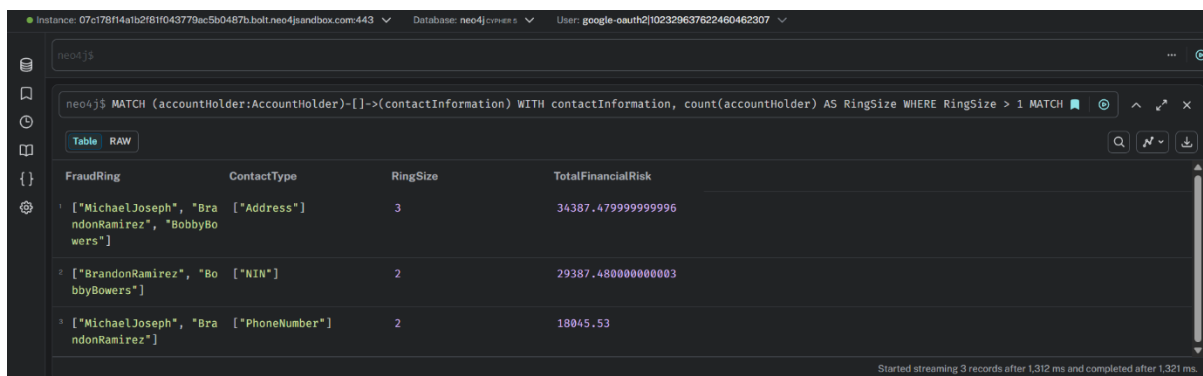
```
MATCH (accountHolder:AccountHolder)-[]->(contactInformation)
WITH contactInformation, count(accountHolder) AS RingSize
WHERE RingSize > 1

MATCH (contactInformation)<-[]-(accountHolder)
OPTIONAL MATCH (accountHolder)-[]->(finance)

WITH contactInformation, collect(DISTINCT accountHolder.UniqueId) AS
FraudRing, RingSize,
sum(
CASE
WHEN finance:CreditCard THEN finance.Limit
WHEN finance:UnsecuredLoan THEN finance.Balance
ELSE 0
END
) AS TotalFinancialRisk

RETURN FraudRing,
labels(contactInformation) AS ContactType,
RingSize,
TotalFinancialRisk
ORDER BY TotalFinancialRisk DESC
```

Output:



FraudRing	ContactType	RingSize	TotalFinancialRisk
["MichaelJoseph", "BrandonRamirez", "BobbyBowers"]	["Address"]	3	34387.479999999996
["BrandonRamirez", "BobbyBowers"]	["NIN"]	2	29387.480000000003
["MichaelJoseph", "BrandonRamirez"]	["PhoneNumber"]	2	18045.53

The query first, finds same contact information nodes connected to multiple account holders and calculates the ring size. It then collects the unique IDs of those accounts that are part of the fraud ring. Then, the query optionally matches financial accounts associated with those account holders and calculate the TotalFinancialRisk which is the total amount of funds (sum of credit card limit and loan balances) that can be potentially stolen by each fraud ring. Finally, it returns the list of account holders in each suspected fraud ring, the type of contact information they share, their ring size, and the total financial risk arranged in descending order (i.e., highest risk to lowest). In this case, three

accounts share the same address to make a fraud ring of size 3 and can cause a total financial risk of 34387.47 (rounded to 2 decimals for reporting).

Lab-3:

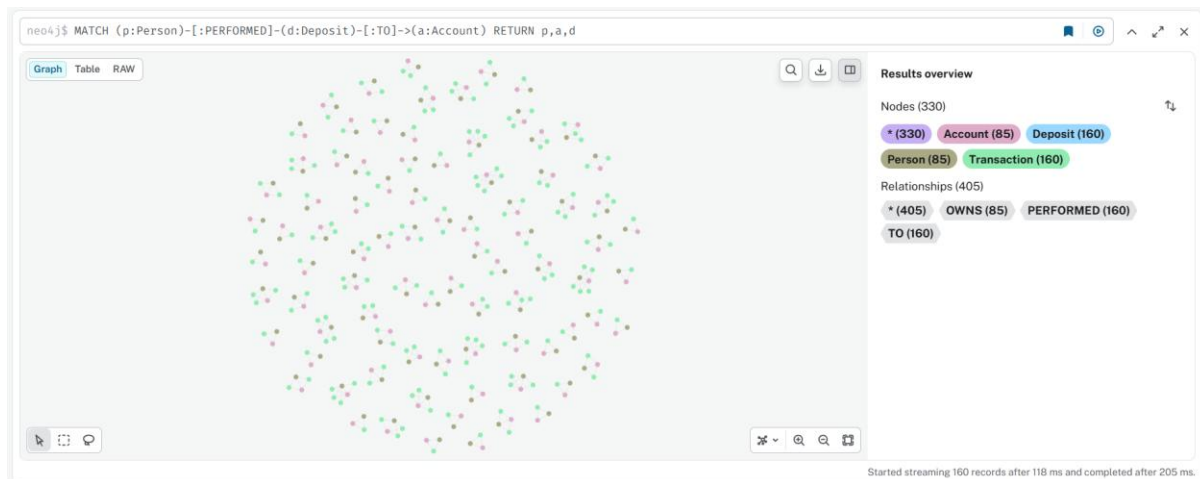
Exercise-1.1: Neo4J temporal analysis using Cypher

The Cypher queries to reach the output:

Query1:

Exploratory analysis was done to the imported data to see if there are some deposits that stand out and create a pattern or some visible outliers.

```
MATCH (p:Person) -[:PERFORMED] - (d:Deposit) -[:TO] -> (a:Account) RETURN p,a,d
```



Query2:

Since nothing looks abnormal, instead of manual checks for each deposit, deposits of same amounts that appear in high frequency is checked since they indicate smurfing activity. This query counts the number of times each deposit amount appears in the dataset and orders them in descending order.

```
MATCH (p:Person) -[:PERFORMED] - (d:Deposit) -[:TO] -> (a:Account)
RETURN d.amount, count(DISTINCT d.amount) AS freq
ORDER BY freq DESC
```

neo4j\$ MATCH (p:Person)-[:PERFORMED]-(d:Deposit)-[:TO]->(a:Account) RETURN d.amount, count(DISTINCT d.amount) AS freq ORDER BY freq DESC

d.amount	freq
3389.7799532883555	1
15951.378147037065	1
7140.822952302806	1
11068.769610198762	1
3694.5842936061104	1
13291.265357774093	1
4810.107247597468	1
13237.703547779014	1
2582.599428274279	1
7303.417664648782	1

Started streaming 160 records after 119 ms and completed after 131 ms.

The output shows that all the deposits have been performed with different amounts so no particular leads.

Query3:

To search for mules, temporal analysis can be done by narrowing the time window (within couple of days), and searching for deposits that sum up to a large amount. To perform mathematical operations on dates, they need to be in integer format so epoch format is used.

```
MATCH (p:Person)-[:PERFORMED]-(d:Deposit)-[:TO]->(a:Account)
WITH min(d.timestamp) AS min_date, max(d.timestamp) AS max_date
RETURN min_date, min_date.epochMillis AS min_date_epoch, max_date, max_date.epochMillis AS max_date_epoch
```

Using the query, the oldest and latest dates of deposits can be identified helping to infer the time window to search in.

neo4j\$ MATCH (p:Person)-[:PERFORMED]-(d:Deposit)-[:TO]->(a:Account) WITH min(d.timestamp) AS min_date, max(d.timestamp) AS max_date

min_date	min_date_epoch	max_date	max_date_epoch
1997-11-21T06:10:32Z	880092632000	2022-02-12T04:31:20Z	1644640280000

Started streaming 1 record after 138 ms and completed after 142 ms.

Query4:

Using the returned time range above, the deposits can be grouped within a sliding window of seven days to see if there is a group depositing a large sum of funds.

```
UNWIND range(880092632000, 1644640280000, 86400000) AS t
MATCH (p:Person)-[:PERFORMED]-(d:Deposit)-[:TO]->(a:Account)
WHERE d.timestamp.epochMillis >= t AND d.timestamp.epochMillis < t + 86400000
RETURN datetime({epochMillis:t}), COUNT(*) as deposit_count
```

```
ORDER BY deposit_count desc
```

neo4j\$ UNWIND range(880092632000, 1644640280000, 86400000) AS t MATCH (p:Person)-[:PERFORMED]-(d:Deposit) WHERE d.timestamp = t

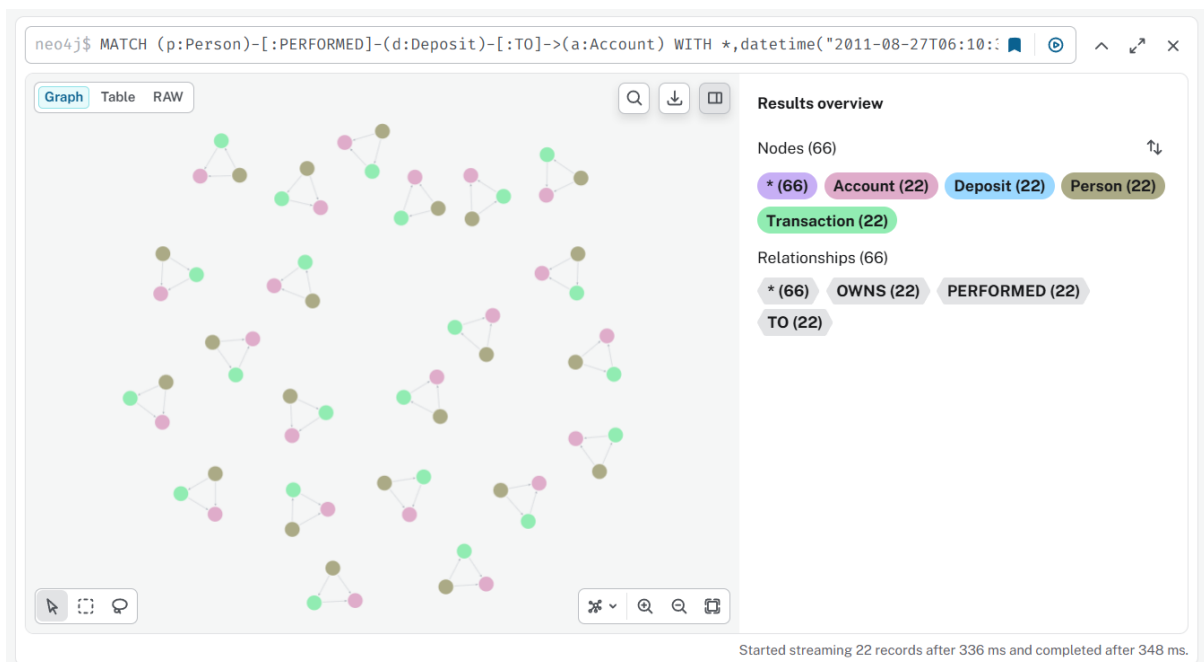
	datetime({epochMillis:t})	deposit_count
1	2011-08-27T06:10:32Z	22
2	2017-08-09T06:10:32Z	15
3	2011-08-28T06:10:32Z	14
4	2017-08-10T06:10:32Z	9
5	2020-07-14T06:10:32Z	2
6	2021-06-04T06:10:32Z	2
7	2022-01-03T06:10:32Z	2
8	1997-11-21T06:10:32Z	1
9	2001-01-27T06:10:32Z	1
10	2001-07-04T06:10:32Z	1

Started streaming 101 records after 605 ms and completed after 4,410 ms.

Query helped us find that there are certain dates where relatively large number of deposits were made.

Query-5:

The date having the most deposits is inspected, and from the properties of the deposits it was seen that the deposits were done with an hour gap.



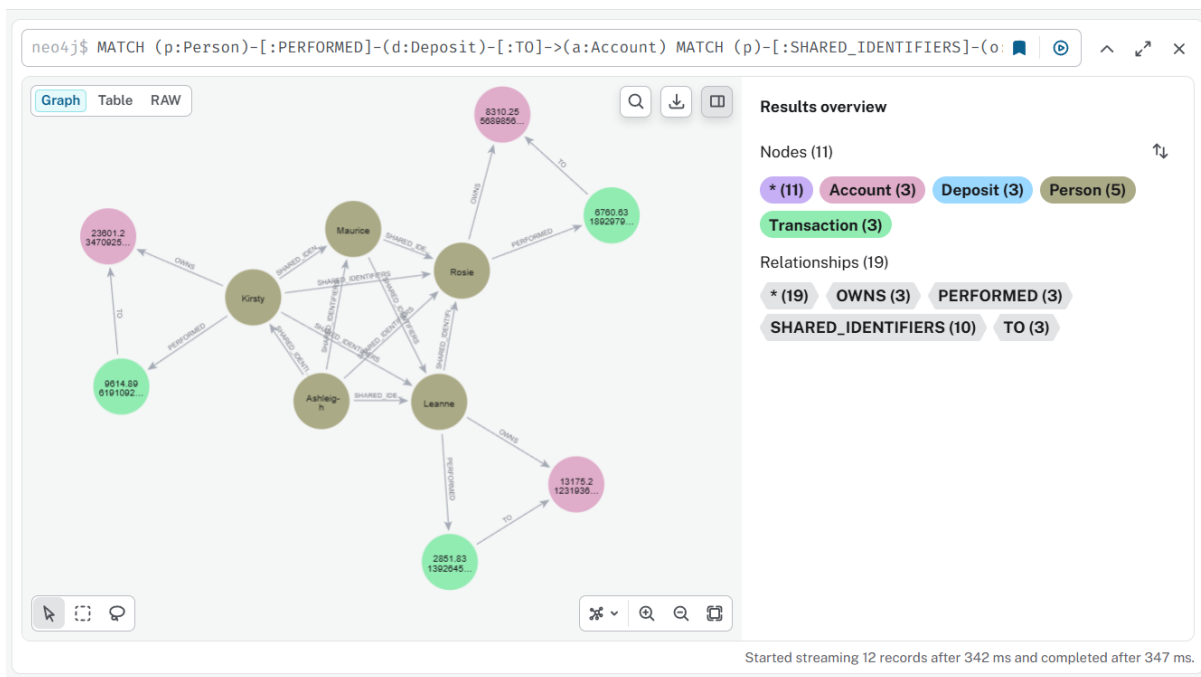
Query-6:

To investigate more and rule out that above was not a coincidence, different bank accounts are checked for personal identifiers shared between them which is typical for fraudsters when creating bank accounts. A new relationship `SHARED_IDENTIFIERS` is created to connect people sharing same information, in this case email and phone.

```
MATCH (p1:Person)-[:HAS_EMAIL|HAS_PHONE]->(info)<-[:HAS_EMAIL|HAS_PHONE]-(p2:Person)
WHERE p1.obj_id<>p2.obj_id
WITH p1, p2, count(*) as cnt
MERGE (p1)-[:SHARED_IDENTIFIERS{count: cnt}]- (p2)
```

Visualizing the relationships using:

```
MATCH (p:Person)-[:PERFORMED]-(d:Deposit)-[:TO]-(a:Account)
MATCH (p)-[:SHARED_IDENTIFIERS]-(o:Person)
WITH *,datetime("2011-08-27T06:10:32Z") as window_start
WHERE d.timestamp.epochMillis >= window_start.epochMillis AND
d.timestamp.epochMillis < window_start.epochMillis + 86400000
RETURN p,d,a,o
```



It is clear that there exists a cluster in the middle which shows people that performed those deposits within the specified time window and on further look they share the same identifiers among them like email and phone information, hinting that they all might be the same person.

Query7:

Query outputs a list of `obj_ids` of suspicious deposits so that they can be reported to the concerned authority.

d.obj_id	
1	"49561b2f-d271-49bf-8a56-0f30f03a69a5"
2	"8294135a-0df4-4310-bf75-3e741a9f00cb"
3	"8294135a-0df4-4310-bf75-3e741a9f00cb"
4	"49561b2f-d271-49bf-8a56-0f30f03a69a5"
5	"c3cefc39-6b93-4831-be31-e34972b1eda2"
6	"8294135a-0df4-4310-bf75-3e741a9f00cb"
7	"c3cefc39-6b93-4831-be31-e34972b1eda2"
8	"49561b2f-d271-49bf-8a56-0f30f03a69a5"
9	"c3cefc39-6b93-4831-be31-e34972b1eda2"
10	"8294135a-0df4-4310-bf75-3e741a9f00cb"
11	"49561b2f-d271-49bf-8a56-0f30f03a69a5"
12	"c3cefc39-6b93-4831-be31-e34972b1eda2"

Started streaming 12 records after 246 ms and completed after 248 ms.

Lab-4:

Exercise-1.1: Neo4J money laundering dataset analysis

Cypher queries to reach the output:

Query1:

Query to perform a simple exploratory analysis where each entity is grouped and counted according to its type.

```
MATCH (n)
RETURN head(labels(n)) as type, count(*) as count
```

neo4j\$ MATCH (n) RETURN head(labels(n)) as type, count(*) as count

Table RAW

type	count
1 "Email"	156
2 "Phone"	155
3 "Person"	159
4 "Trust"	16
5 "ShellCompany"	27
6 "Bank"	10
7 "Account"	143
8 "Transaction"	1160
9 "Asset"	1

Started streaming 9 records after 24 ms and completed after 28 ms.

The output shows quite a lot of data for each entity so exploring them manually is impractical.

Query2:

To start the investigation, a metric called in-degree of a node is calculated which shows the number of relationships that point to that node. So high in-degree shows more incoming relationships (high incoming transactions) suggesting suspicious activity when it is performed in a short period. This query will provide all the in and out degrees of the Account nodes in our graph as they are our main entities that perform transactions. Here, all the Account nodes are matched and degree of a node is obtained with the help of apoc.node.degree from APOC procedure. "<" represents in -degree, whereas ">" represents out-degree. They are arranged in descending order according to inDegree value as it is the driving metric for the investigation.

```
MATCH (p:Account)
WITH apoc.node.degree(p,"<") AS inDegree, apoc.node.degree(p,">") AS outDegree
RETURN inDegree, outDegree
ORDER BY inDegree DESC;
```


neo4j\$ MATCH (p:Account) WITH apoc.node.degree(p,"<") AS ir

Table RAW

	inDegree	outDegree
1	22	2
2	20	4
3	20	18
4	19	10
5	18	14
6	18	14
7	18	12
8	17	10
9	17	15
10	17	9

Started streaming 143 records after 59 ms and completed after 66 ms.

Query3:

This query is similar to previous but this time the search is more specific for accounts that receive money ("<TO" relationship) through many transactions, ordered from the most to lowest(descending).

```
MATCH (p:Account)
RETURN apoc.node.degree(p,"<TO") AS inDegree
ORDER BY inDegree DESC;
```

neo4j\$ MATCH (p:Account) RETURN apoc.node.degree(p,"<TO")

Table RAW

	inDegree
1	21
2	19
3	19
4	18
5	17
6	17
7	17
8	16
9	16
10	16

Started streaming 143 records after 62 ms and completed after 65 ms.

Output showed that out of 21 out of 22 incoming relationships mapped to money transfers, raising the suspicions more.

Query4:

To confirm the suspicions, the timespan for the transactions is examined. Using this query, the obj_id of the subject node who had such amount of inDegree is obtained so that it can be further investigated.

```
MATCH (p:Account)
RETURN apoc.node.degree(p,"<TO") AS inDegree, p.obj_id
ORDER BY inDegree DESC;
```

neo4j\$ MATCH (p:Account) RETURN apoc.node.degree(p,"<T0") AS inDeg

Table RAW

	inDegree	p.obj_id
1	21	"38f2a17e-88a9-45ee-acab-8dbfd125241e"
2	19	"eb2b40b6-2276-4071-8700-44b153b3d02c"
3	19	"b3a00e8a-457b-4afc-b860-d11cf8309628"
4	18	"171a434e-97fd-4eeb-a6f0-aa950eb87112"
5	17	"da6df937-c734-4b81-bb55-cb074c27368d"
6	17	"73205398-474a-40f0-af70-7d2498e5f706"
7	17	"6436b154-81b6-425b-a220-65df65c72d97"
8	16	"649deed4-f81b-49cf-a2ae-19b52581c968"
9	16	"6b52cf50-b7f7-4f6f-8ccc-cb874b5e39c5"
10	16	"f29458a8-300d-40a2-abd7-0714b51af86b"

Started streaming 143 records after 32 ms and completed after 35 ms.

Query5:

Now, using this query, the minimum and maximum dates of the transfers can be identified as well as the period in-between them in hours for that specific obj_id ("38f2a17e-88a9-45ee-acab-8dbfd125241e"). Again, the time is in epoch format for convenience, but since we want to know the time window in hours it is divided by 1000, 60 and 60 so that milliseconds convert to hours.

```
MATCH (p:Account) <-[:TO]-(t:Transfer)
WHERE p.obj_id = "38f2a17e-88a9-45ee-acab-8dbfd125241e"
WITH min(t.timestamp) as minDate, max(t.timestamp) as maxDate
WITH *, (maxDate.epochMillis-minDate.epochMillis)/1000/60/60 as
hoursTimeDelta
RETURN minDate, maxDate, hoursTimeDelta
```

```

1 MATCH (p:Account)<-[:TO]-(t:Transfer)
2 WHERE p.obj_id = "38f2a17e-88a9-45ee-acab-8dbfd125241e"
3 WITH min(t.timestamp) as minDate, max(t.timestamp) as maxDate
4 WITH *, (maxDate.epochMillis-minDate.epochMillis)/1000/60/60 as
  hoursTimeDelta
5 RETURN minDate, maxDate, hoursTimeDelta

```

minDate	maxDate	hoursTimeDelta
2021-03-02T11:19:00Z	2021-03-03T07:17:00Z	19

Started streaming 1 record after 118 ms and completed after 120 ms.

The output actually shows that all of the transfers happened within the 19 hours which suggests further investigation.

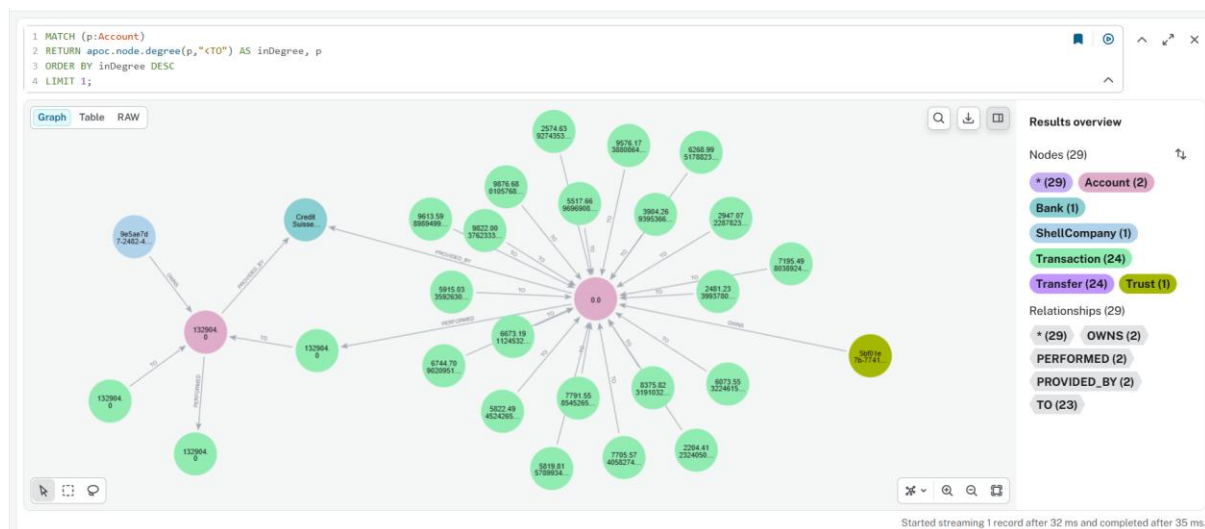
Query6:

This query shows the visualization of all the relationships of the node of interest by limiting the initial degree command to return only the node with the highest in-degree value using LIMIT 1.

```

MATCH (p:Account)
RETURN apoc.node.degree(p,"<TO") AS inDegree, p
ORDER BY inDegree DESC
LIMIT 1;

```



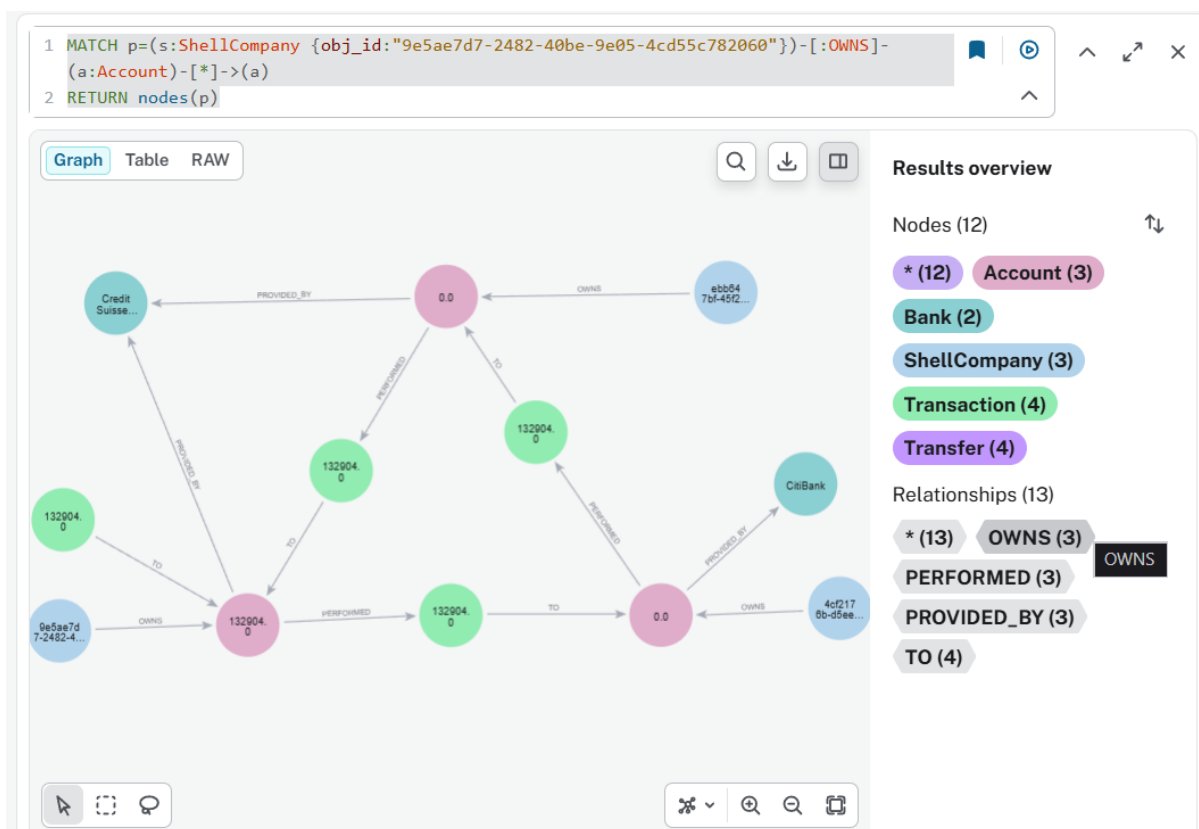
Expanding the node further it was visible that the account was owned by a Trust and it had performed an outgoing transaction almost 1.5 hour after the last incoming transaction showcasing a strong intent to transfer all the money received to a single account. Also, expanding the transaction further

showed that the money was transferred to an account owned by a Shell company. All of these points towards the hypothesis that they are a part of Layering activity.

Query7:

The query investigates for round tripping i.e., it matches obj_id of the Shell company in question to search for accounts from where the funds are transferred from and to so that they end up back to the source. Further expanding the graph showed that there actually was roundtripping between 3 distinct accounts and they were all owned by shell companies.

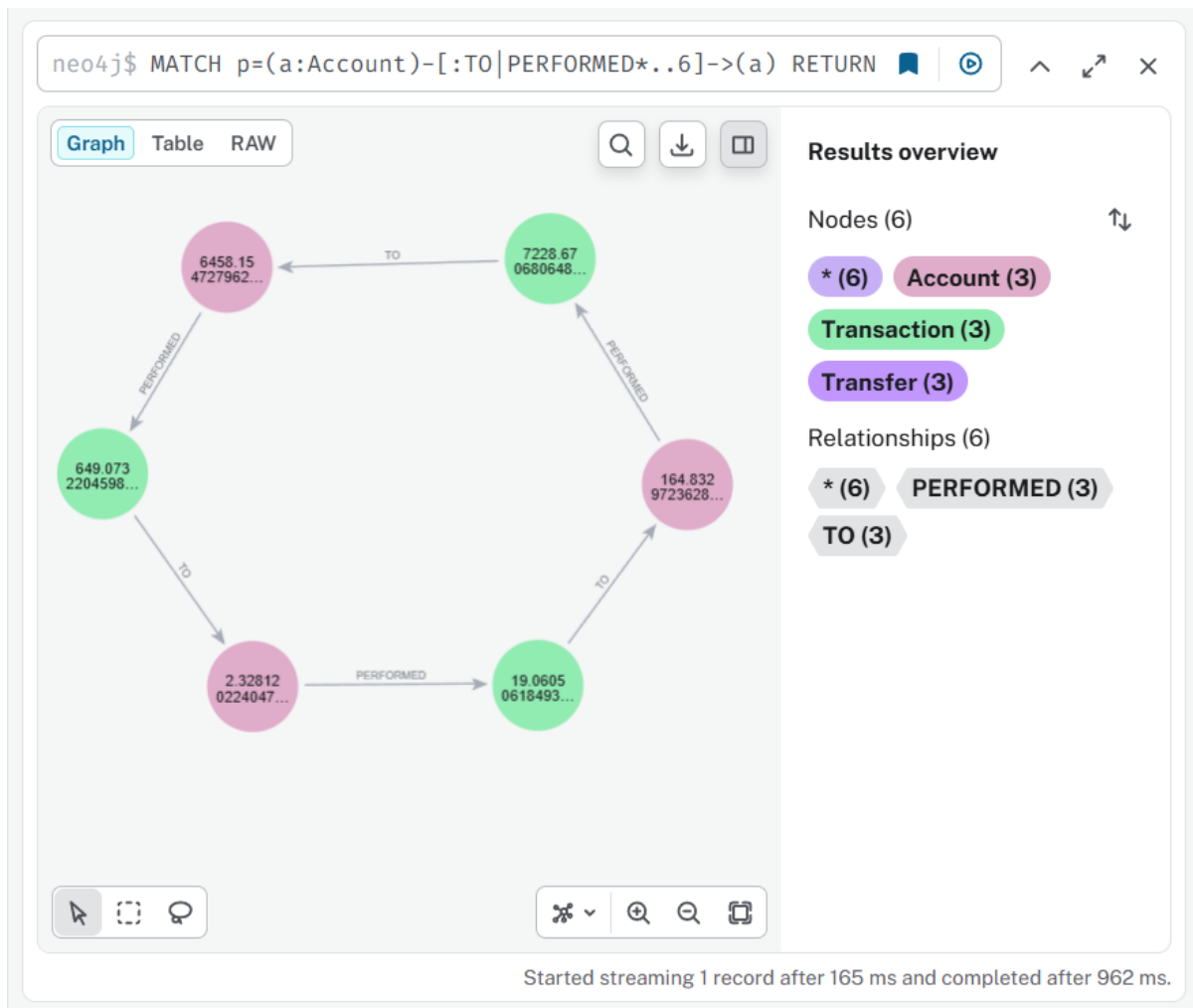
```
MATCH p=(s:ShellCompany {obj_id:"9e5ae7d7-2482-40be-9e05-4cd55c782060"})-[:OWNS]-(a:Account)-[*]->(a)
RETURN nodes(p)
```



Query8:

This query just searches for round trips across the whole graph with some constraints for faster querying.

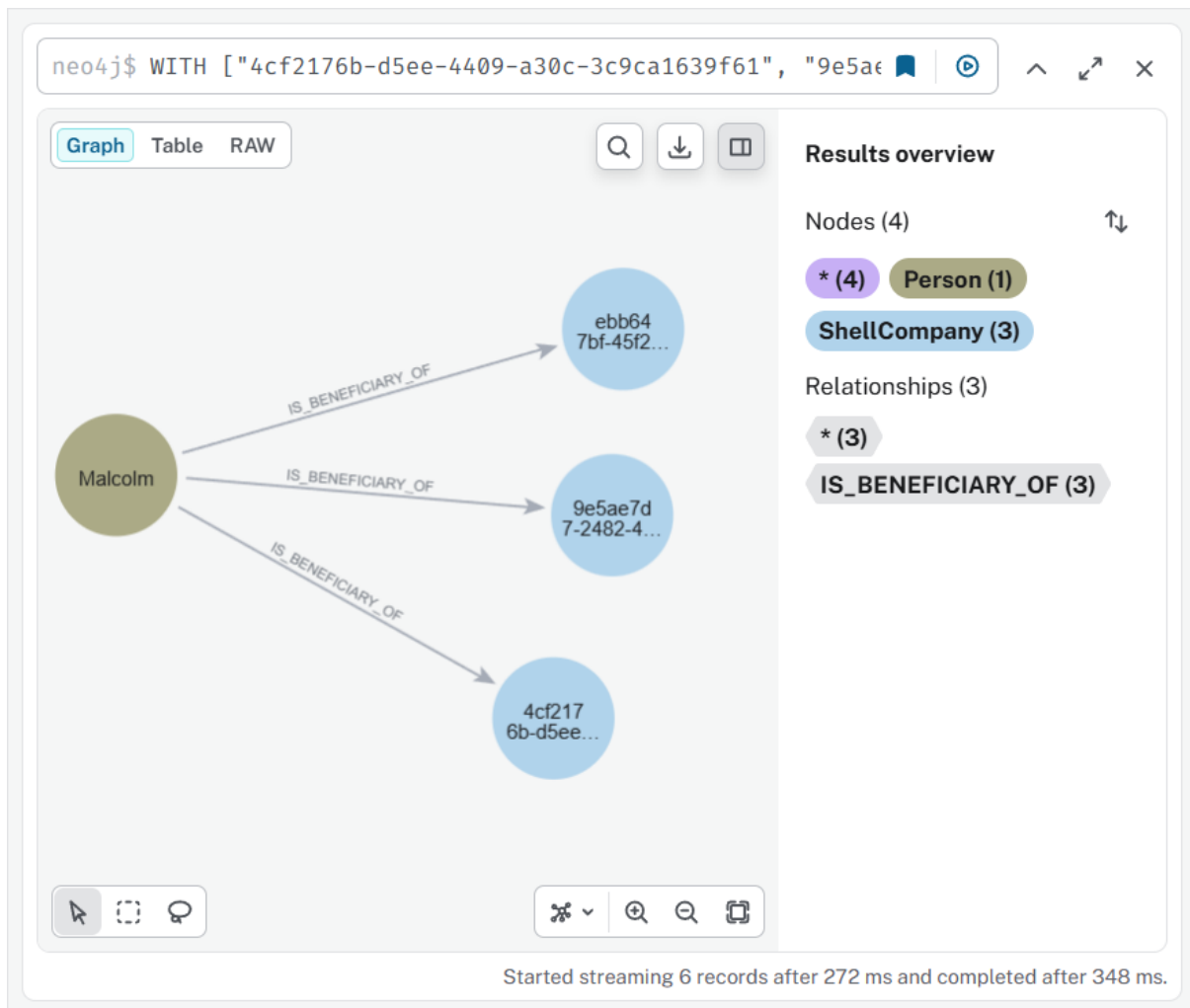
```
MATCH p=(a:Account)-[:TO|PERFORMED*..6]->(a)
RETURN p
ORDER BY length(p) DESC
LIMIT 1
```



Query9:

This query uses the shell companies' ids that were obtained during investigation of round tripping to find any first-degree connections that link those together while limiting the relationships to a single hop connection to avoid intractable queries.

```
WITH ["4cf2176b-d5ee-4409-a30c-3c9ca1639f61", "9e5ae7d7-2482-40be-9e05-4cd55c782060", "ebb647bf-45f2-4246-868d-4ed9d2a30449"] as
shellCompaniesIds
MATCH p=(s1:ShellCompany)-[*..1]-()-[*..1]-(s2:ShellCompany)
WHERE s1.obj_id IN shellCompaniesIds AND s2.obj_id IN
shellCompaniesIds AND
s1.obj_id <> s2.obj_id
RETURN p
```

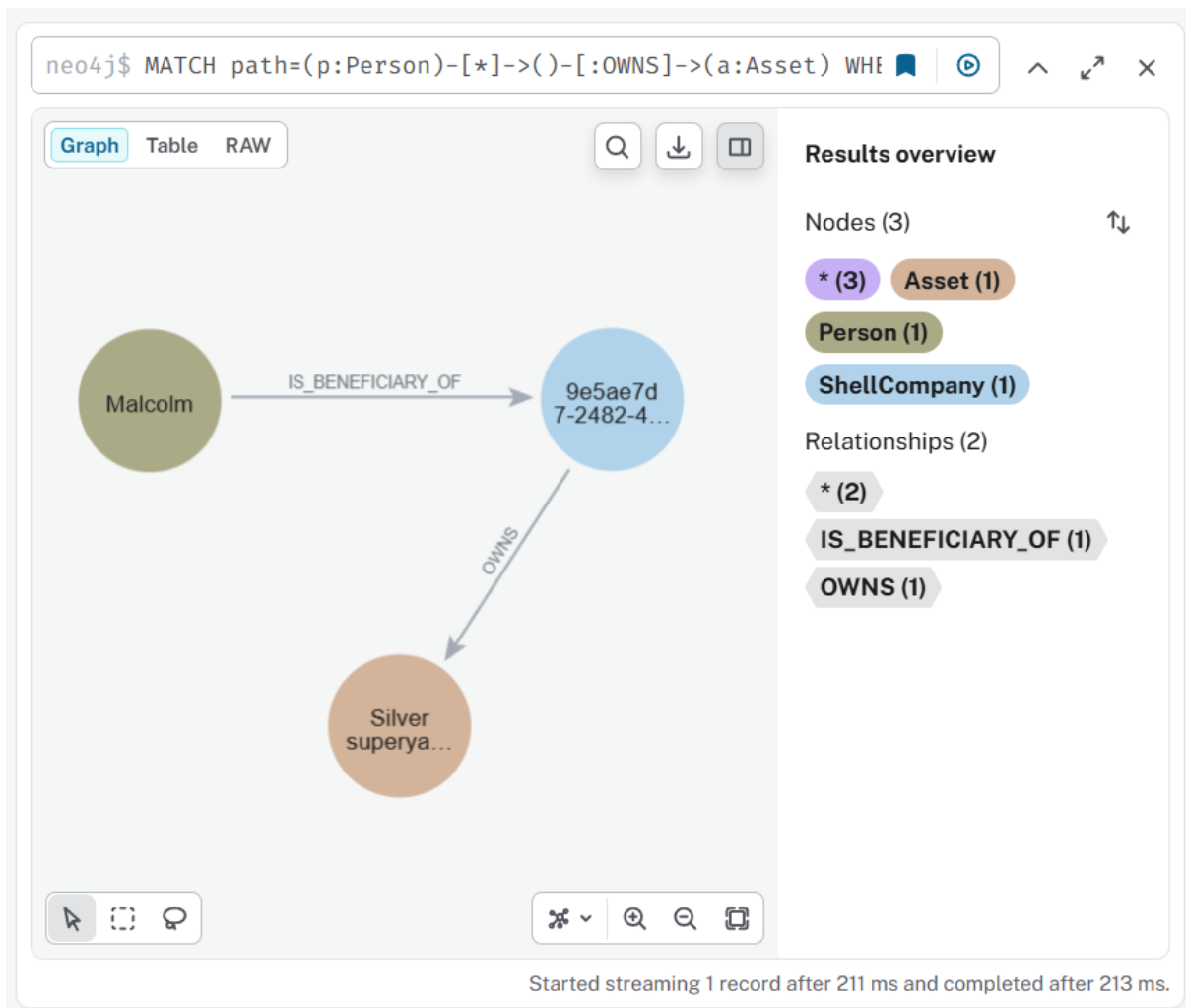


The output shows that these shell companies all have the same beneficiary with the name “Malcolm Brown” which fuels the suspicions of involvement in round tripping for layering.

Query10:

This query takes the investigation further by querying for any indirect connection from this Person to a high valuable asset which represents the Integration phase. Here the search is for assets having valuation more than 10000.

```
MATCH path=(p:Person)-[*]->()-[:OWNS]->(a:Asset)
WHERE p.obj_id = "580fc4ad-8b88-4f4b-9ef1-19ea974eafa3" AND
toFloat(a.value_in_gbp) > 10000
RETURN path
```

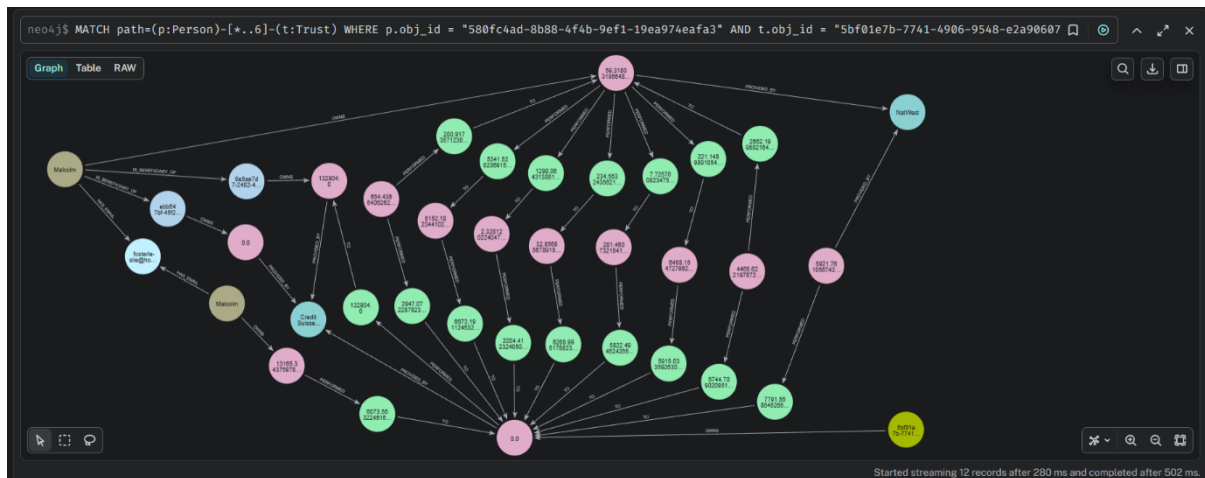


Output shows that indeed that Malcolm owns a high valuable physical asset i.e., Silver superyacht through one of his shell companies. This confirms that he has been involved in money laundering.

Query11:

This query shows the whole picture of money laundering lifecycle through all stages querying for any relationships (up to 6th degree) that connects the Person to the Trust:

```
MATCH path=(p:Person)-[*..6]-(t:Trust)
WHERE p.obj_id = "580fc4ad-8b88-4f4b-9ef1-19ea974eafa3" AND t.obj_id =
"5bf01e7b-7741-4906-9548-e2a90607f4c5"
RETURN path
```

Final Output:

Hence based on the analysis, the obj_id and name of the person involved in money laundering is:

obj_id: "580fc4ad-8b88-4f4b-9ef1-19ea974eafa3"

Name: "Malcolm Brown"

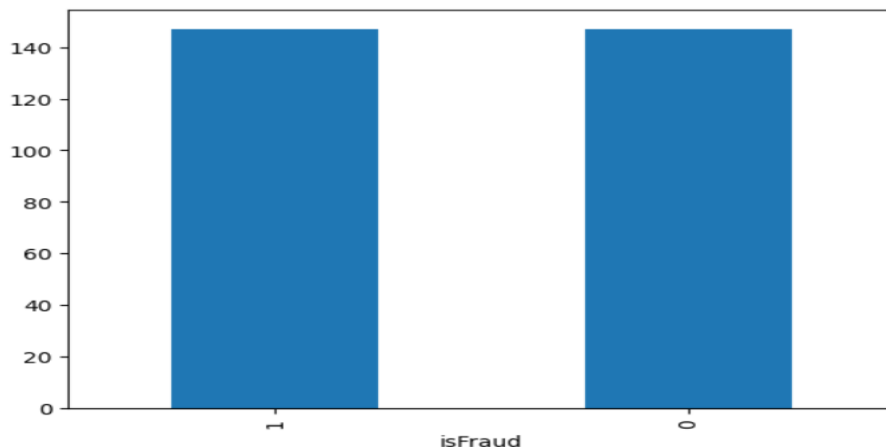
Lab-5:

The python notebooks are zipped alongside the report. Some plots from the notebook are shown here.

Exercise-1.4(a): Balance an imbalanced dataset (Refer to notebook LabExercise5.ipynb for full code and plots)

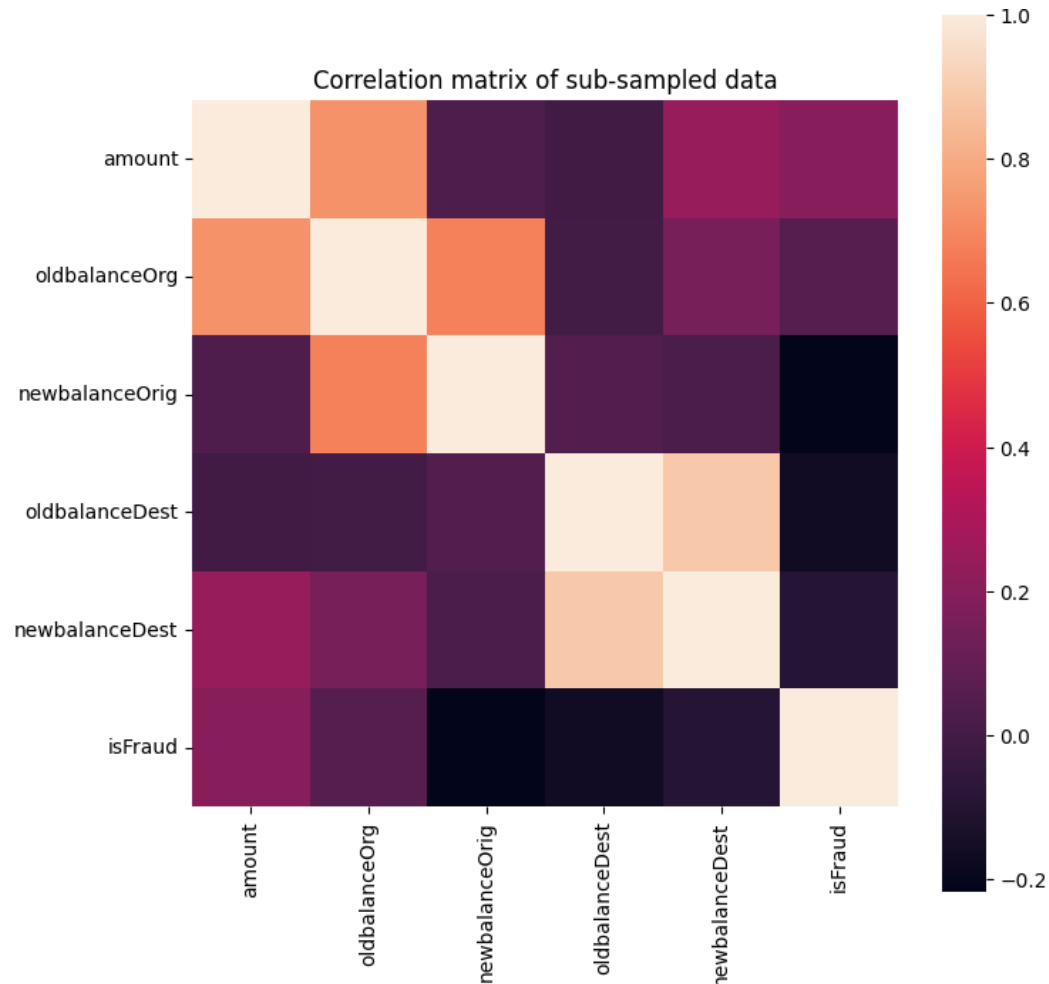
The dataset was balanced using Undersampling technique, which involves removing samples from the training dataset that belong to the majority class in order to better balance the class distribution.

```
... <Axes: xlabel='isFraud'>
```



Exercise-1.4 (b): Correlation heatmap (Refer to notebook LabExercise5.ipynb for full code and plots)

Correlation heatmaps helps to identify features that are closely correlated with the legitimacy of our transactions. This helps in feature selection.

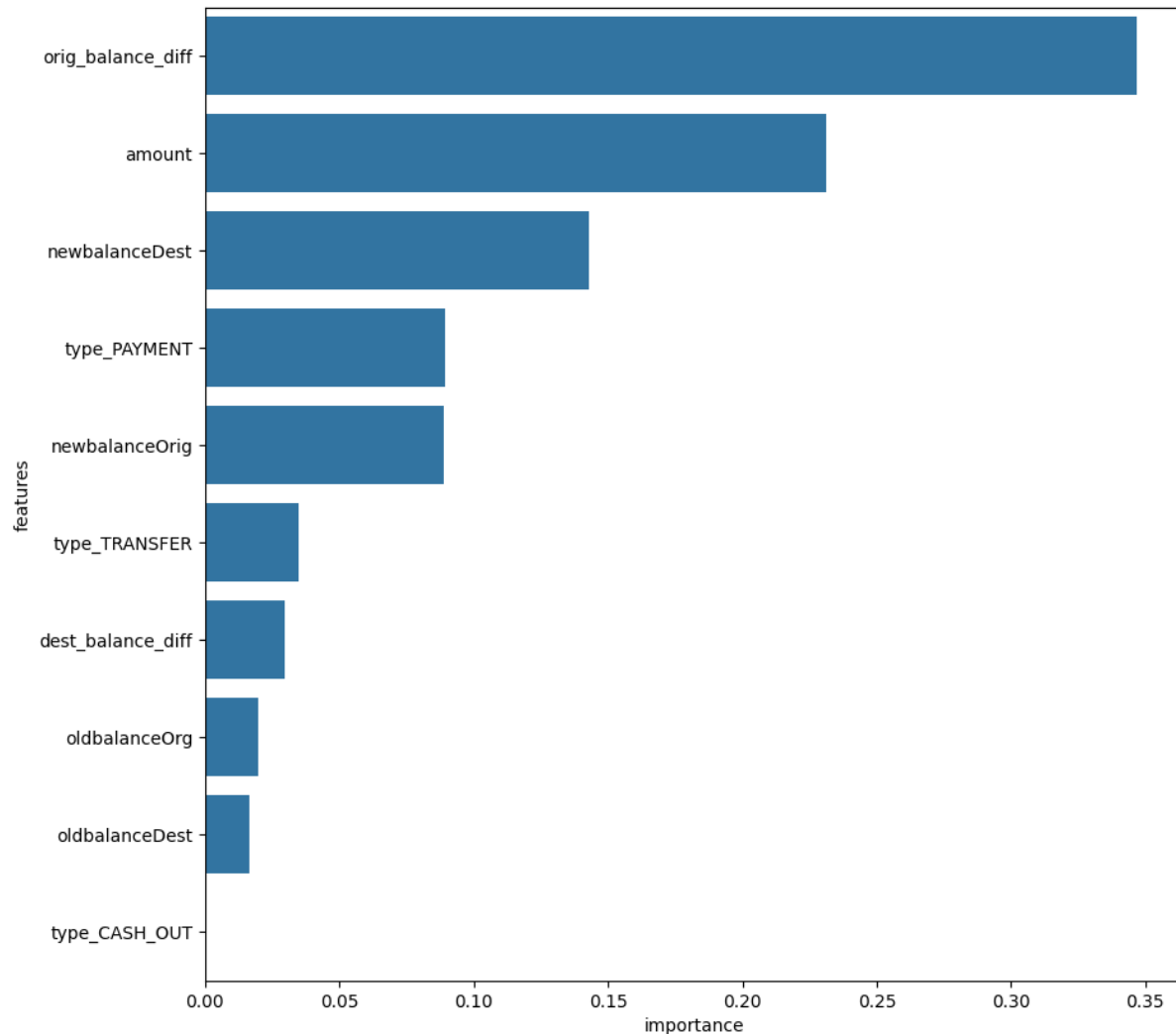


Exercise-1.8: Create decision tree classifier (Refer to notebook LabExercise5.ipynb for full code and plots)

A decision tree model works by sequentially splitting the dataset along value thresholds of the predictor variables.

Exercise-1.10: Feature importance (Refer to notebook LabExercise5.ipynb for full code and plots)

Feature importance helps to identify which feature influence the predictions outcome of the model most. This can be obtained using Sklearn's `feature_importances_` attribute of `DecisionTreeClassifier`.



Feature importance plot shows the feature that was made when performing feature engineering 'orig_balance_diff' as the most important feature for the model, with importance value of 0.346811.

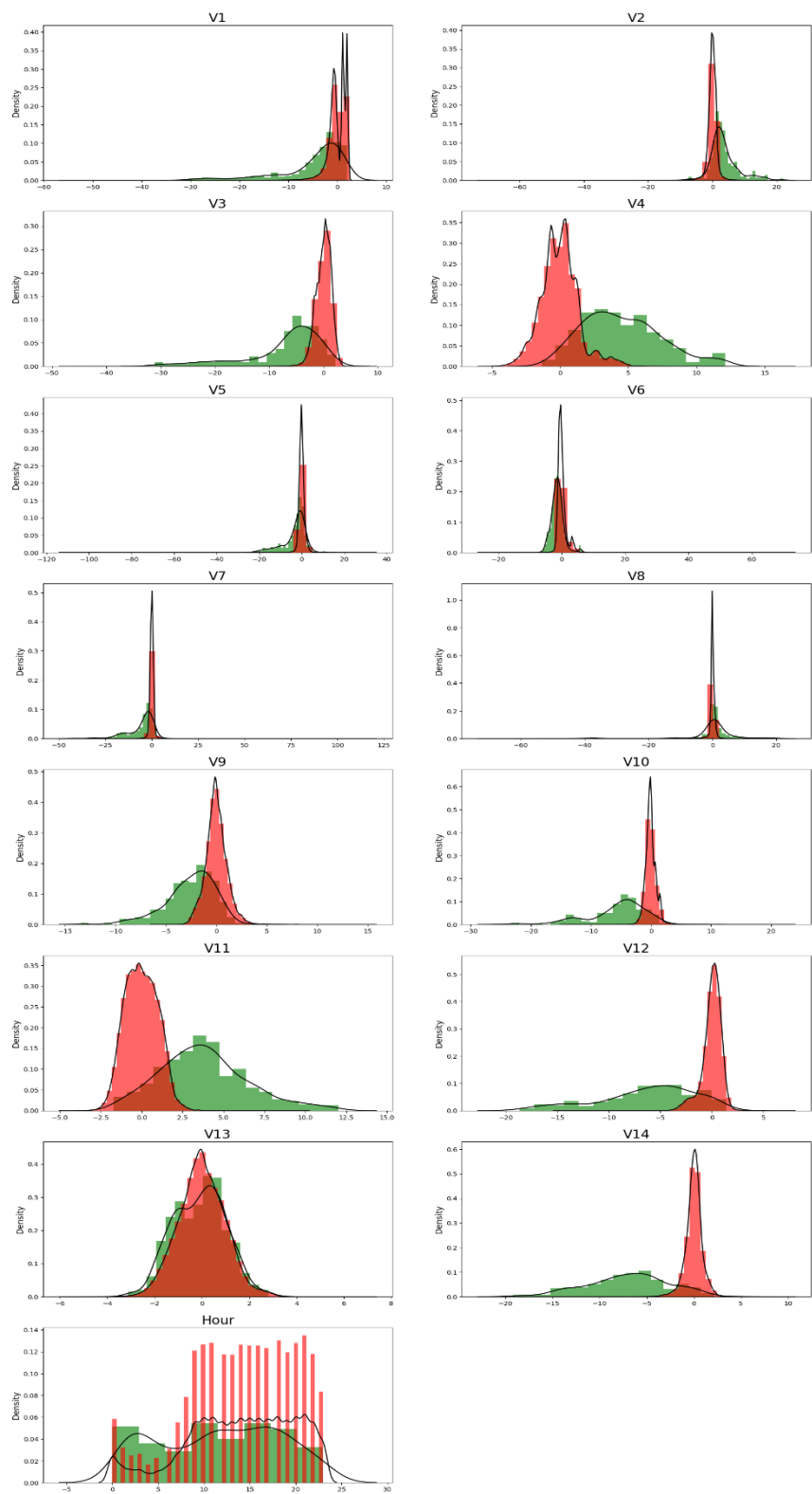
It is the difference between old and new balance of the original account obtained by:

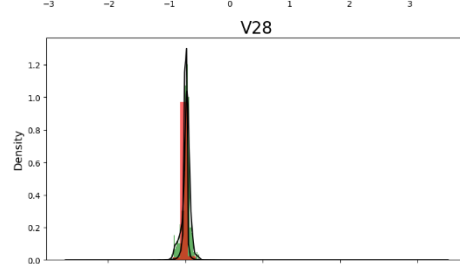
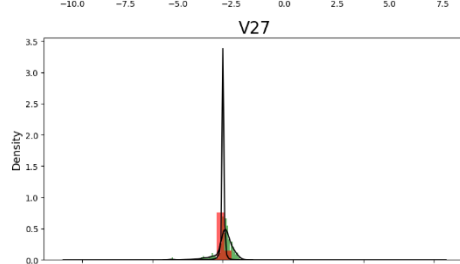
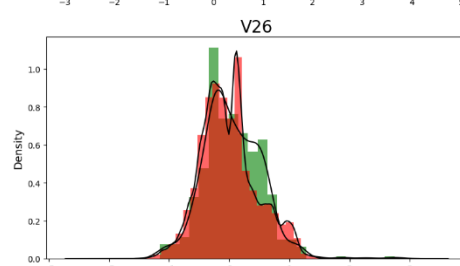
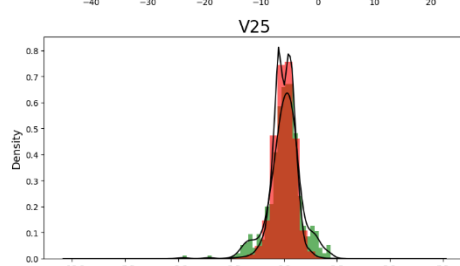
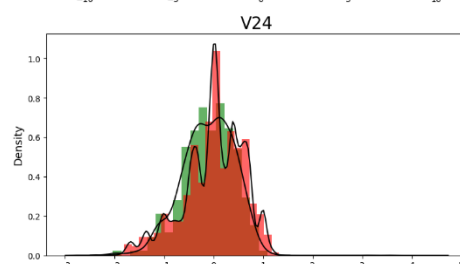
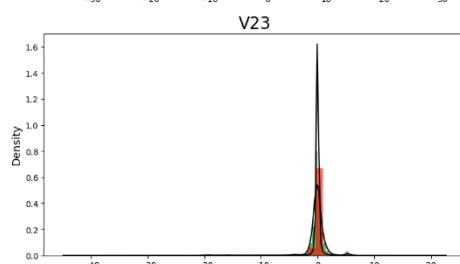
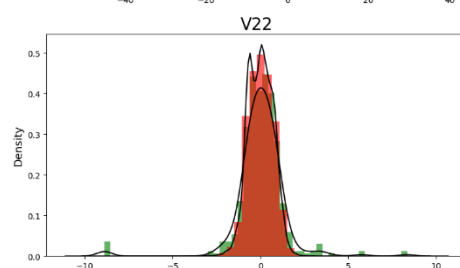
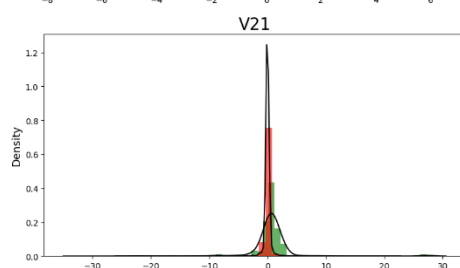
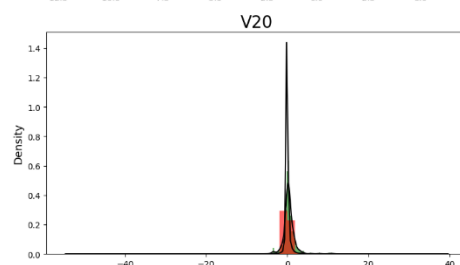
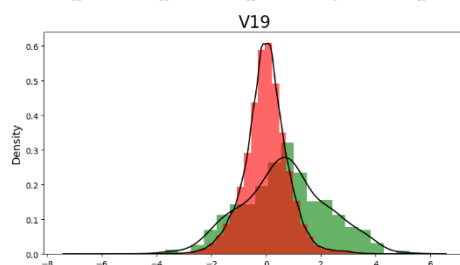
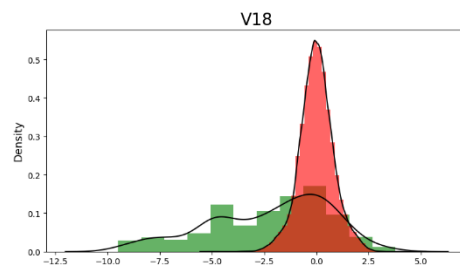
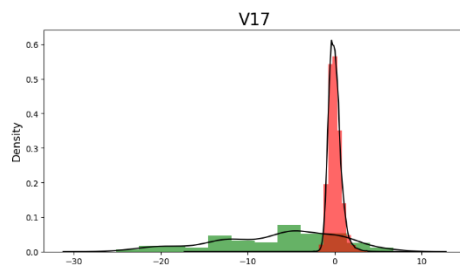
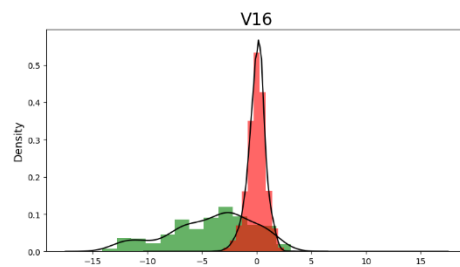
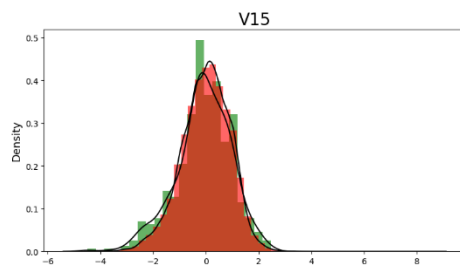
```
balanced_df['orig_balance_diff'] = balanced_df['oldbalanceOrg'] - balanced_df['newbalanceOrig']
```

This shows the importance of feature engineering and how a very useful fraud feature was engineered from existing features.

Lab-6:

Exercise-3.1: K-Means clustering (Refer to notebook LabExercise6.ipynb for full code and plots)





Based on the KDE plots, I selected the feature pair: 'V14' and 'V17' because for clustering algorithms like K-Means clustering, they work best when there is clear class separability i.e., it works best when groups can be naturally differentiated into distinct regions in the feature space.

V14 and V17 plots both show they have clear separation between fraudulent and non-fraudulent transactions i.e., a noticeable difference between two classes. This separation indicates that both of them possess strong discriminative power so it can help in distinguishing fraudulent from legitimate transactions. Also, both features have classes where there is tight cluster of non-fraudulent ones and wide spread of fraudulent transactions allowing the algorithm to form more meaningful clusters.

Additionally, there are some variables like V13, V20, V21 and many more show heavy overlap between the classes which causes difficulty in natural grouping of transactions.

The prediction accuracy for the selected feature was **72.499%**.

... 72.499%					
	precision	recall	f1-score	support	
0	1.00	0.72	0.84	284315	
1	0.00	0.76	0.01	492	
accuracy			0.72	284807	
macro avg	0.50	0.74	0.42	284807	
weighted avg	1.00	0.72	0.84	284807	

SMOTE is an oversampling technique where it generates synthetic samples for the minority class by clustering the data using KNN algorithm and sampling new data on the lines that join the minority class points.

```
Original dataset shape : Class
0      284315
1         492
Name: count, dtype: int64
Resampled dataset shape Class
0      284315
1      284315
Name: count, dtype: int64
```

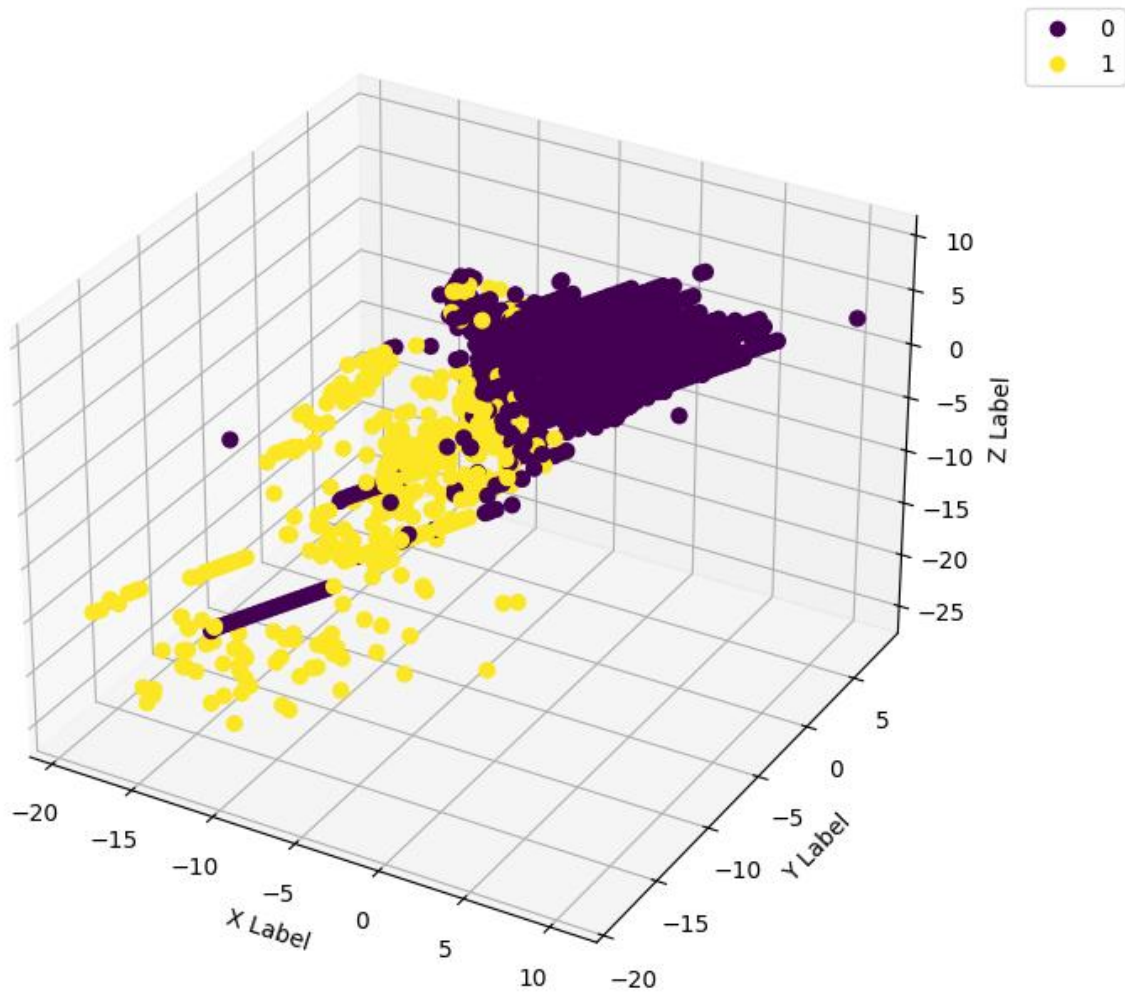
Using SMOTE on the dataset and re-running K-means clustering improved the prediction accuracy slightly to give **74.846%**.

*** 74.846%

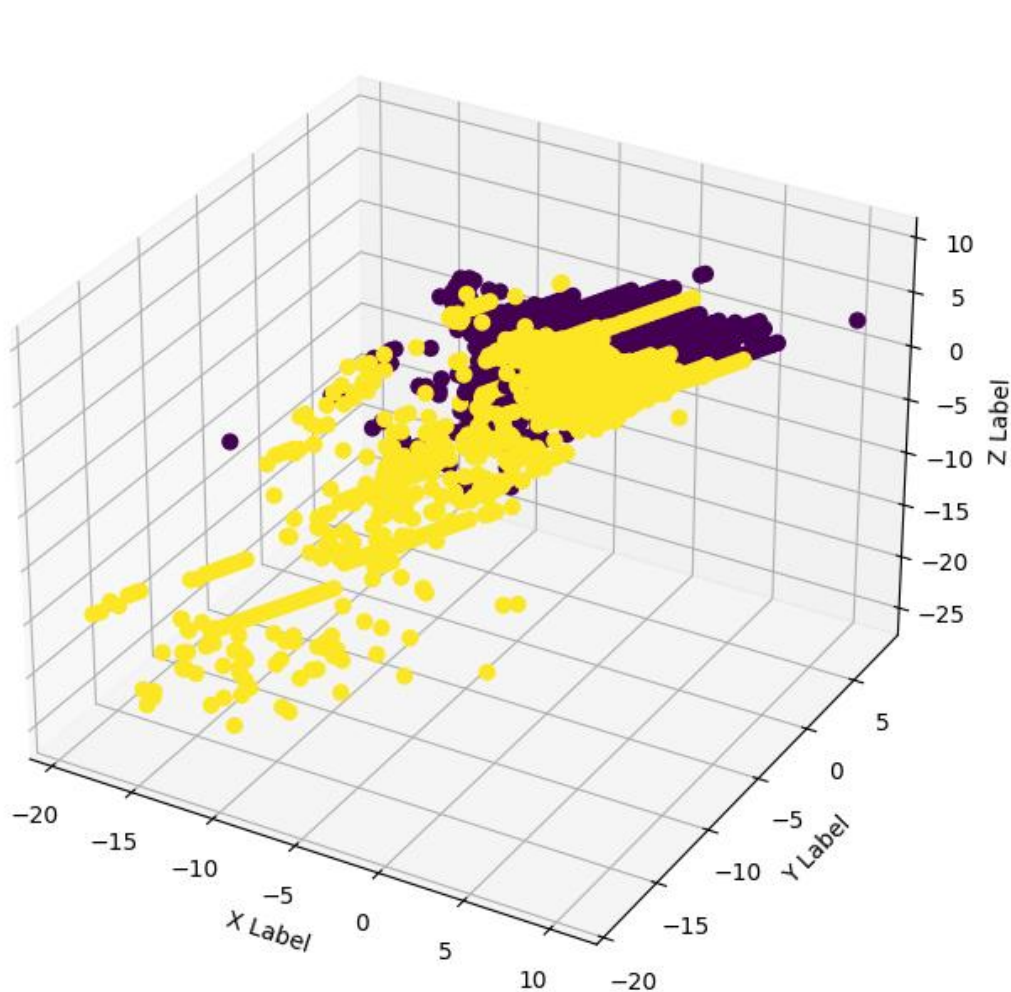
	precision	recall	f1-score	support
0	0.67	1.00	0.80	284315
1	1.00	0.50	0.66	284315
accuracy			0.75	568630
macro avg	0.83	0.75	0.73	568630
weighted avg	0.83	0.75	0.73	568630

Exercise-3.2: 3D Scatterplot (Refer to notebook LabExercise6.ipynb for full code and plots)

To plot the 3d scatterplot in 3d space, three input features 'V14' and 'V12' and 'V17' were selected. When the data points are color coded by using "Class" column, following plot was obtained:



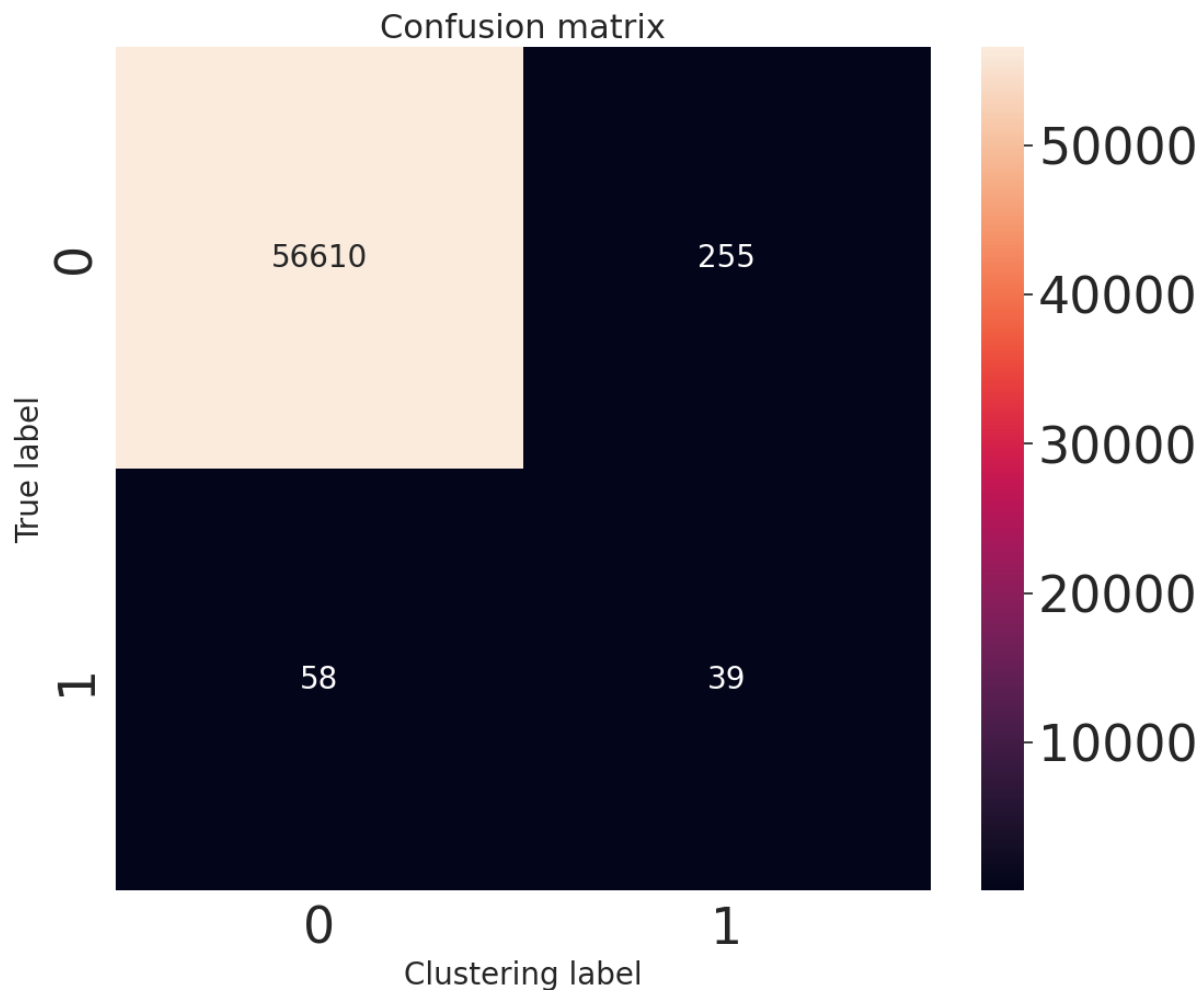
After performing K-means clustering on these 3 features and plotting them using K-means cluster values for color coding gives:



Exercise-3.3: Isolation Forest (Refer to notebook LabExercise6.ipynb for full code and plots)

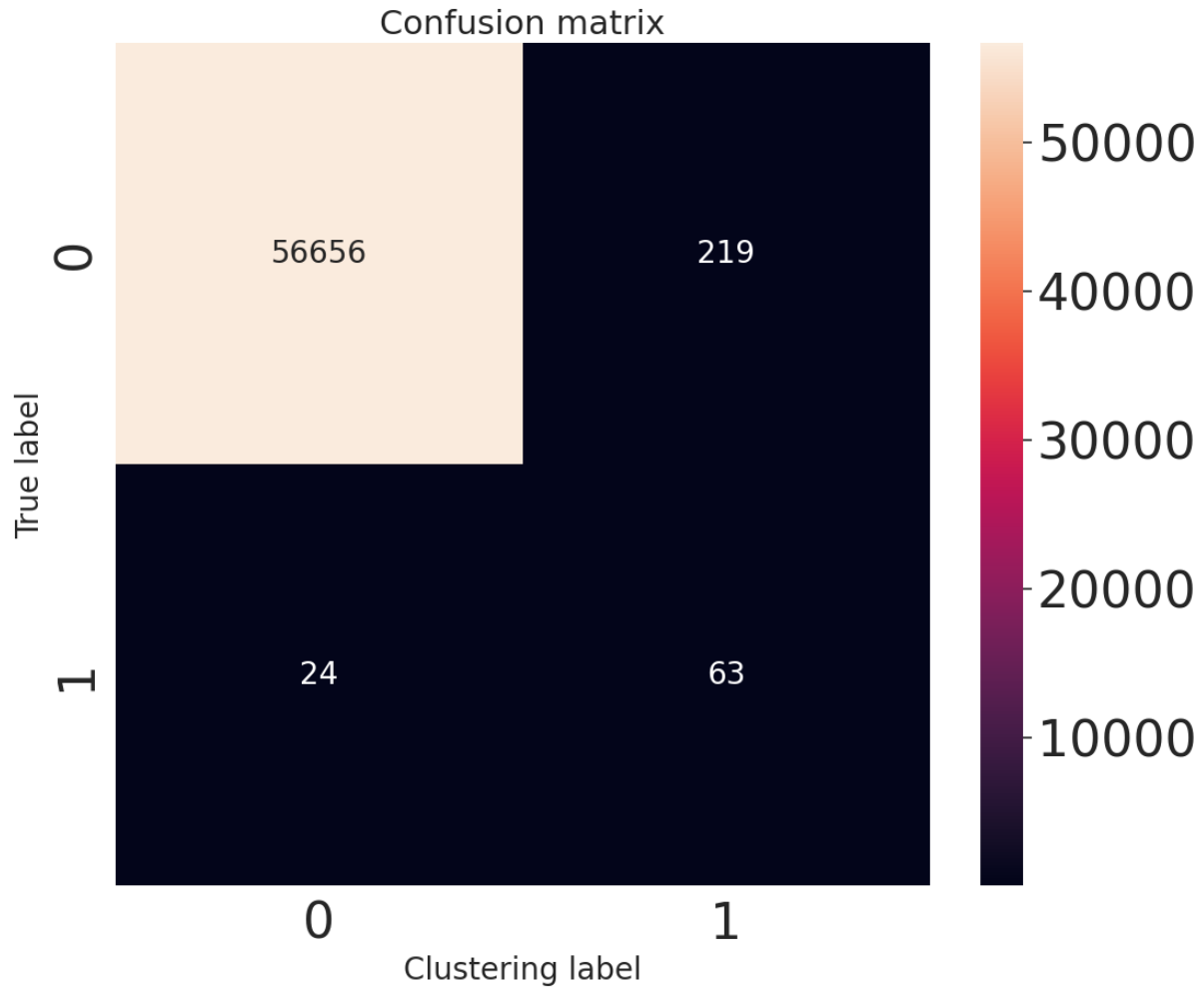
The example's Isolation Forest model uses features from V1 to V9 and obtains an accuracy of **0.99** but macro avg is just **0.60**.

	precision	recall	f1-score	support
0	1.00	1.00	1.00	56865
1	0.13	0.40	0.20	97
accuracy			0.99	56962
macro avg	0.57	0.70	0.60	56962
weighted avg	1.00	0.99	1.00	56962



Selecting different subset of features: 'V1', 'V2', 'V3', 'V4', 'V11', 'V12', 'V14', 'V16' and 'V17', and re-running the Isolation Forest algorithm gives an improved performance of accuracy **1** and a much better macro avg of **0.67**.

	precision	recall	f1-score	support
0	1.00	1.00	1.00	56875
1	0.22	0.72	0.34	87
accuracy			1.00	56962
macro avg	0.61	0.86	0.67	56962
weighted avg	1.00	1.00	1.00	56962



Lab-7:

Exercise-1.1: Use blockchain explorer to track transactions

After opening the “Sepolia” testnet Etherscan blockchain explorer and searching for the given wallet address “0xA80ACBcA81235411D856628698323079ddFe9Fd6” we observe the following:

Address 0xA80ACBcA8123411D856628698323079ddFe9Fd6

Overview
ETH BALANCE
0.0045 ETH

More Info
TRANSACTIONS SENT
Latest: N/A First: N/A
FUNDED BY
0xB0f17Fd1...49cE6A68f | 1 yr 2 days ago

Multichain Info
N/A

Transactions Token Transfers (ERC-20)

Latest 1 from a total of 1 transactions

Transaction Hash	Method	Block	Age	From	To	Amount	Txn Fee
0x27a87ebf44c...	Transfer	7928442	367 days ago	0xB0f17Fd1...49cE6A68f	0xA80ACBcA...9ddFe9Fd6	0.0045 ETH	0.00011281

<https://sepolia.etherscan.io/address/0xB0f17Fd183A47114aB955F4C3a6fF4e49cE6A68f>

In the Transaction section, clearly it shows from which address the funds were transferred from or FUNDED BY section as well to show the wallet address which is found to be: “0xB0f17Fd183A47114aB955F4C3a6fF4e49cE6A68f”

Again, after backtracking further to find the funds origin on that wallet as well, we find that the above wallet was funded by: “0x67a2210AaB20912ea072F61a22d7f6C6B995B4cA”

Address 0xB0f17Fd183A47114aB955F4C3a6fF4e49cE6A68f

Overview
ETH BALANCE
0.000387189362291 ETH

More Info
TRANSACTIONS SENT
Latest: 1 yr 2 days ago First: 1 yr 2 days ago
FUNDED BY
0x67a2210A...6B995B4cA | 1 yr 2 days ago

Multichain Info
N/A

Transactions Token Transfers (ERC-20)

Latest 2 from a total of 2 transactions

Transaction Hash	Method	Block	Age	From	To	Amount	Txn Fee
0x27a87ebf44c...	Transfer	7928442	367 days ago	0xB0f17Fd1...49cE6A68f	0xA80ACBcA...9ddFe9Fd6	0.0045 ETH	0.00011281
0xff0f62da6bb...	Transfer	7928439	367 days ago	0x67a2210A...6B995B4cA	0xB0f17Fd1...49cE6A68f	0.005 ETH	0.00011227

<https://sepolia.etherscan.io/address/0x67a2210AaB20912ea072F61a22d7f6C6B995B4cA>

Hence, the two most recent addresses that those funds originate from are:

Address 1 : 0xB0f17Fd183A47114aB955F4C3a6fF4e49cE6A68f

Address 2 : 0x67a2210AaB20912ea072F61a22d7f6C6B995B4cA

Exercise-1.2: Use blockchain explorer to explain transactions

BNB Price: \$652.42 (+0.09%) Gas: 0.1 Gwei

Search by Address / Txn Hash / Block / Token / Domain Name

TRANSACTION ACTION

Swap 1,181,668.513145 Mattie for 0.164338052957493228 WBNB (\$107.05) on PancakeSwap

Transaction Hash: 0x4aafcd1e3fcc175024e27f40abe0dfd62a9256f8e8181c8426c674b45b499624

Status: Success

Block: 11145250 7505680 Block Confirmations

Timestamp: 1631 days ago (Sep-22-2021 08:00:58 PM UTC)

Sponsored: **TRADE PERPS**
Crypto, commodities, stocks, and more

From: 0xc6Fd786bB7416ece05e6691eE4967ad062393842

Interacted With (To): 0x10ED43C718714eb63d5aA57B78B54704E256024E (PancakeSwap: Router v2)

Internal Transactions:

All Transfers **Net Transfers**

- Transfer 0.164338052957493228 BNB \$107.22 From BNB Chain: WBNB Token To PancakeSwap: Router v2
- Transfer 0.164338052957493228 BNB \$62.30 From PancakeSwap: Router v2 To 0xc6Fd786b...062393842

BEP-20 Tokens Transferred: 2

All Transfers **Net Transfers**

- 0xc6Fd786b...062393842 sent 1,181,668.513145 BEP-20: Mattie (Mattie)
- PancakeSwap: Router v2 received 0.164338052957493228 (\$107.06) Wrapped BNB (WBNB)
- PancakeSwap V2: Matt... received 1,181,668.513145 BEP-20: Mattie (Mattie)
- PancakeSwap V2: Matt... sent 0.164338052957493228 (\$107.06) Wrapped BNB (WBNB)

Value: 0 BNB (\$0.00)

Transaction Fee: 0.00058986 BNB \$0.38

Gas Price: 6 Gwei (0.000000006 BNB)

The user initiated a successful token swap executed through PancakeSwap. The wallet “0xc6Fd786bB7416ece05e6691eE4967ad062393842” interacted with the PancakeSwap Router v2 to trade 0.164338052957493228 WBNB for 1,181,668.513145 Mattie. During the swap, the WBNB was sent from the user's wallet to the PancakeSwap liquidity pool through the router contract, and in return the liquidity pool transferred Mattie tokens back to the user's wallet. This swap took a small transaction fee of 0.00058986 BNB.

Exercise-3.1: Anonymity set of mixer transactions sample

What is the anonymity set of the pool at deposit D5?

The Anonymity Set is defined by the total number of deposits among which an observer cannot distinguish a specific transaction.

Anonymity set at deposit D5 = 2

Explanation: At Deposit D1 the Anonymity Set becomes 1 then it is followed by a withdrawal of W1. This reduces the Anonymity set back to 0 as the withdrawal at that time can be directly mapped to that single deposit. In this way, at D2 it becomes 1 then 0 again due to withdrawal W2. Deposits D3 and D4 before D5 makes the anonymity set to 2 and the anonymity set of a certain deposit at that time is noted for the value before that deposit. Hence, it is 2.

What is the anonymity set of the pool at withdrawal W6?

Anonymity set at withdrawal W6 = 1

Explanation: Continuing from before, it is known that at D5 the anonymity set was 2 then deposit D5 makes it 3. After that it is followed by consecutive withdrawals W4 and W5 which reduces the anonymity set back to 1 before another withdrawal of W6. Hence, it is 1.

Given the anonymity set at time of withdrawal W6, is there any anonymity added to the withdrawal? Explain your answer.

No there is no anonymity added to the withdrawal. It's because the level of anonymity provided at the time of withdrawal is directly related to the amount of time that the funds remain in the pool. It's like hiding within the crowd but for anonymity to be added, new deposits must enter the mixer between deposit and withdrawal timeframe.

Since at the time of W6, there is only one deposit left in the pool that has not been withdrawn, the withdrawal can only correspond to that one remaining deposit leaving the privacy of the remaining funds at its lowest value. This makes W6 as the last person, making it easier to link the withdrawal to the remaining original deposit.

Lab-8:

Exercise-1.1: Deanonimisation of wallets used in mixers

The cypher queries to reach the output:

Query1:

The following query computes the anonymity set for each transaction by processing them in chronological order and calculates two values for each transaction: open_count and anonymity_set. It iterates through the transactions one by one using apoc.periodic.iterate and finds the most recent previous transaction to determine the current state. If a deposit happens, both open_count and anonymity_set are increased, indicating more deposits entering the mixer. Similarly, if a withdrawal occurs, open_count decreases (but never less than zero). The anonymity_set stays the same during the withdrawals and only resets to zero when open_count becomes zero, meaning the mixer pool has been fully drained.

```
CALL apoc.periodic.iterate(
  "MATCH (t:Transaction)
  WHERE t.method_id IN ['deposit', 'withdraw']
  RETURN t
  ORDER BY t.timestamp ASC",
  "WITH t
  OPTIONAL MATCH (prevTx:Transaction)
  WHERE prevTx.timestamp < t.timestamp
  AND prevTx.method_id IN ['deposit', 'withdraw']
  WITH t, prevTx
  ORDER BY prevTx.timestamp DESC
  LIMIT 1
  WITH t,
  prevTx,
  CASE
```

```

WHEN prevTx IS NULL AND t.method_id = 'deposit' THEN 1
WHEN prevTx IS NULL AND t.method_id = 'withdraw' THEN 0
WHEN t.method_id = 'deposit' THEN coalesce(prevTx.open_count, 0) + 1
WHEN t.method_id = 'withdraw' THEN
CASE
WHEN coalesce(prevTx.open_count, 0) > 0
THEN prevTx.open_count - 1
ELSE 0
END
END AS newOpenCount
SET t.open_count = newOpenCount,
t.anonymity_set =
CASE
WHEN t.method_id = 'deposit'
THEN coalesce(prevTx.anonymity_set, 0) + 1
WHEN t.method_id = 'withdraw' THEN
CASE
WHEN newOpenCount = 0 THEN 0
ELSE coalesce(prevTx.anonymity_set, 0)
END
END
END
",
{batchSize: 1, parallel: false}
)

```

batch	total	time	comm	failed	failed	retrie	errorMessage	batch	operations	wasT	failedParam	updateStat
1	77	77	1	77	0	0	0	{}	{ total:77, committed:77, failed:0, errors:{} }	false	{}	{ nodesDeleted:0, labelsAdded:0, relationsCreated:0, nodesCreated:0, propertiesSet:154, relationsDeleted:0, labelsRemoved:0 }

Started streaming 1 record after 27 ms and completed after 1.841 ms.

Query2:

This query is for low anonymity withdrawals, identifying possible deposit sources for each withdrawal transaction. It first finds withdrawals sent to the mixer and then looks for the most recent transaction where the anonymity set reset to zero, indicating the mixer pool was previously drained. Then, using this reset point as a boundary, the query selects all deposits to the same mixer that occurred after the last reset and before withdrawing. Then, a POSSIBLE_DEPOSIT relationship between the withdrawal and the deposits is created which represents the set of deposits that could potentially be the source of the withdrawn funds.

```

MATCH (w:Transaction) -[:TO]-> (c:Contract)
WHERE w.method_id = 'withdraw'

```

```

OPTIONAL MATCH (prevReset:Transaction)-[:TO]->(c)
WHERE prevReset.timestamp < w.timestamp
AND prevReset.anonymity_set = 0
WITH w, c, prevReset
ORDER BY prevReset.timestamp
DESC WITH w, c, collect(prevReset)[0] AS lastReset
MATCH (d:Transaction)-[:TO]->(c)
WHERE d.method_id = 'deposit'
AND d.timestamp > coalesce(lastReset.timestamp, 0)
AND d.timestamp <= w.timestamp
MERGE (w)-[:POSSIBLE_DEPOSIT]->(d);

```

Query3:

This query counts the number of possible deposits linked to each withdrawal transaction. It matches withdrawals and optionally finds all the POSSIBLE_DEPOSIT relationships connected to them. Then the number of deposits is counted and stored in the property possible_deposit_count for each withdrawal.

```

MATCH (w:Transaction)
WHERE w.method_id = 'withdraw'
OPTIONAL MATCH (w)-[:POSSIBLE_DEPOSIT]->(d:Transaction)
WITH w, count(d) AS cnt
SET w.possible_deposit_count = cnt;

```

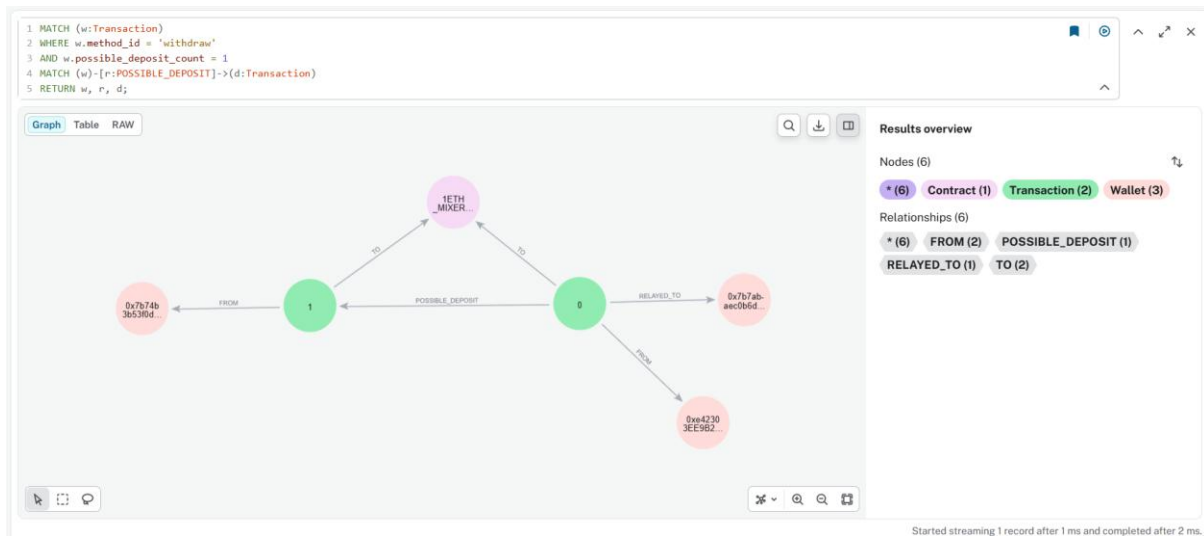
Query4:

This query identifies withdrawals that have exactly one possible deposit source. It matches withdrawals where possible_deposit_count = 1 and then, retrieves the single POSSIBLE_DEPOSIT relationship and the corresponding deposit transaction. The result shows the withdrawal, the relationship and the deposit which is the only possible source of the withdrawn funds.

```

MATCH (w:Transaction)
WHERE w.method_id = 'withdraw'
AND w.possible_deposit_count = 1
MATCH (w)-[r:POSSIBLE_DEPOSIT]->(d:Transaction)
RETURN w, r, d;

```

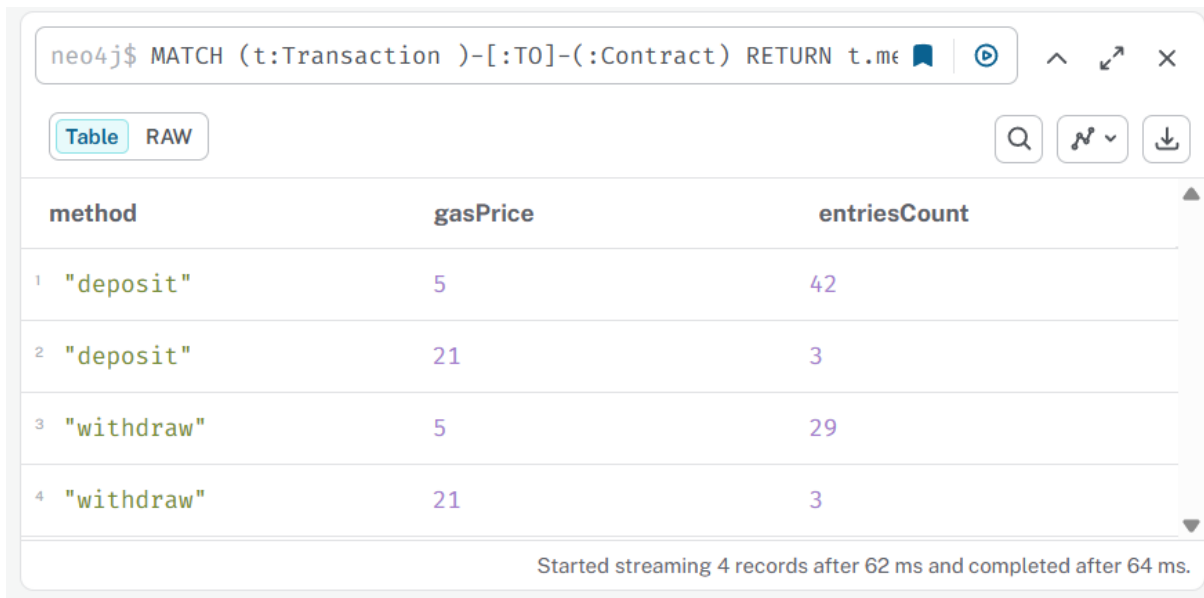


Hence, from the output it can be seen that the address: "0x7b74b3b5f0d651ceb56206607e0fbc9dd247d" is connected to address: "0x7b7abaec0b6d1b4704ed624cc96d464d397faa67". This is because the query produces a graph view identifying the withdrawal that has exactly one possible source deposit, anonymity set effectively reduced to a single candidate, significantly reducing privacy.

Query5:

The query below utilises the gas price property of the transactions to draw relationships between wallet. It displays all gas prices aggregated by the transaction method_id ("deposit", "withdraw").

```
MATCH (t:Transaction )-[:TO]-(:Contract)
RETURN t.method_id AS method, t.gas_price AS gasPrice, COUNT(*) AS
entriesCount
ORDER BY method, gasPrice
```

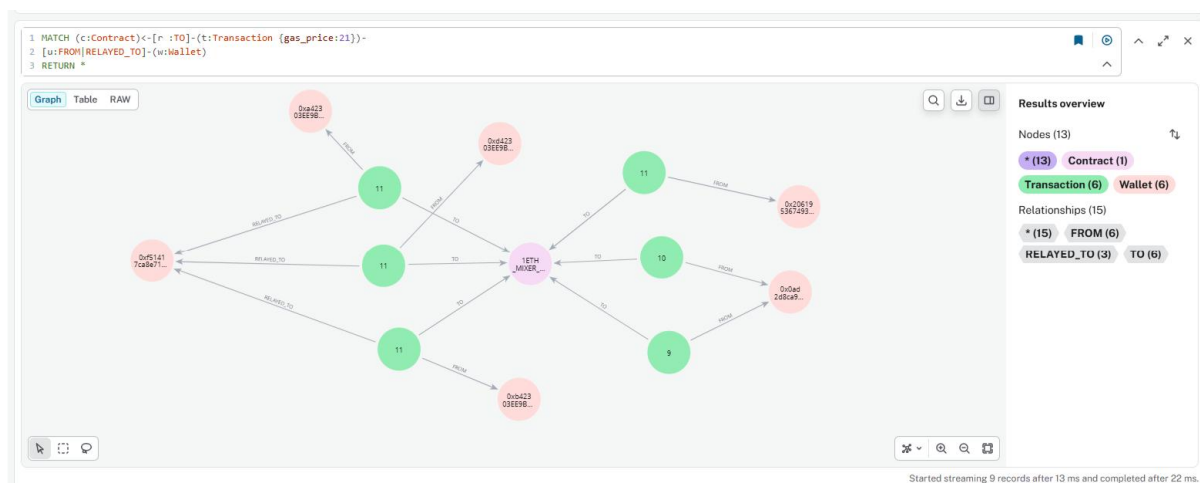


The deposit and withdraw costing gasPrice 21 and entriesCount 3 look to specific to be a coincidence. This can be taken as a strong indication that they are potentially related.

Query6:

This query is to visualise the transactions with a gas price of 21 that interact with a mixer contract. It matches the TO relationship between the transaction and the contract, and then identifies the wallet connected to the transaction through either a FROM or RELAYED_TO relationship.

```
MATCH (c:Contract) <- [r :TO] - (t:Transaction {gas_price:21}) -
[u:FROM|RELAYED_TO] - (w:Wallet)
RETURN *
```



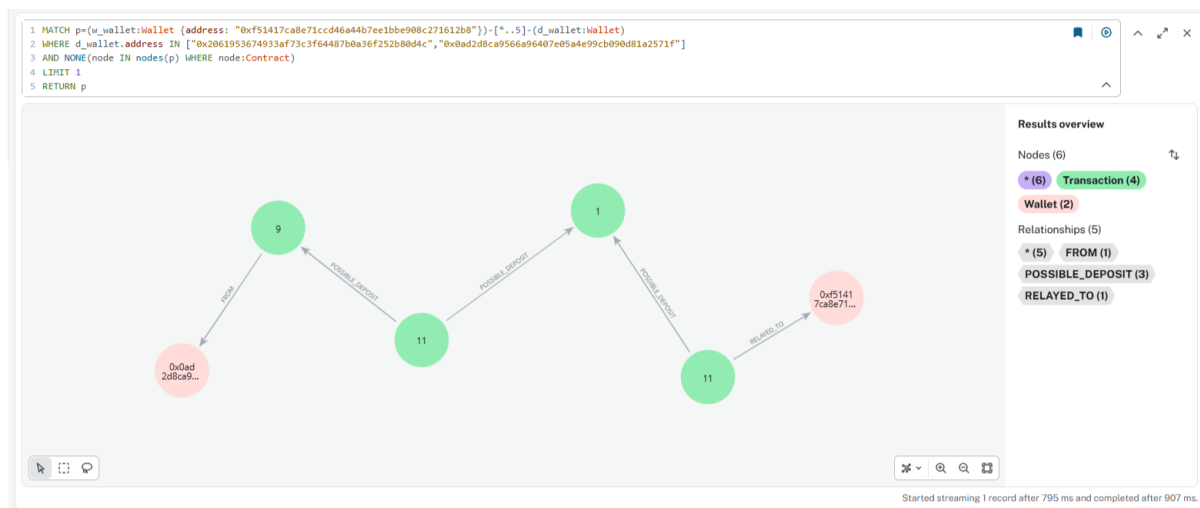
Hence, from the output above, deposit address: "0x2061953674933af73c3f64487b0a36f252b80d4c", deposit address: "0x0ad2d8ca9566a96407e05a4e99cb090d81a2571f" and withdrawal address: "0xf51417ca8e71ccd46a44b7ee1bbe908c271612b8" are connected to each other. All of their transactions whether withdraw or deposit cost the same gas_price of 21 which is very suspicious.

Query7:

This query is for linked address matching where the public blockchain data is utilised to uncover any other links between the obtained addresses. It finds a path to the withdrawal address "0xf51417ca8e71ccd46a44b7ee1bbe908c271612b8" from any of the deposit addresses: "0x2061953674933af73c3f64487b0a36f252b80d4c" and "0x0ad2d8ca9566a96407e05a4e99cb090d81a2571f", but not through the mixer. It queries for 5 hops(*..5) and restricted to only a single path (LIMIT 1).

```
MATCH p=(w_wallet:Wallet {address:
"0xf51417ca8e71ccd46a44b7ee1bbe908c271612b8"}) - [*..5] -
(d_wallet:Wallet)
WHERE d_wallet.address IN
["0x2061953674933af73c3f64487b0a36f252b80d4c", "0x0ad2d8ca9566a96407e05
a4e99cb090d81a2571f"]
AND NONE (node IN nodes(p) WHERE node:Contract)
```

```
LIMIT 1
RETURN p
```



From the output graph, it is quite clear that the deposit address: "0x0ad2d8ca9566a96407e05a4e99cb090d81a2571f" and withdrawal address: "0xf51417ca8e71ccd46a44b7ee1bbe908c271612b8" are connected as they had funds transferred between them. The withdraws and deposits are also done by the same person using the same wallet which pretty much explains the connection between them.

Exercise-1.2: Anonymity set of the pool

The current anonymity set of mixer pool based on the data provided is **22**. The following query retrieves the most recent anonymity set value from the transaction dataset. It matches all transactions where the method is either deposit or withdraw, orders them by timestamp in descending order, and returns the anonymity_set of the latest transaction. The LIMIT 1 ensures that only the most recent value is returned.

Query to obtain the output:

```
MATCH (t:Transaction)
WHERE t.method_id IN ['deposit', 'withdraw']
RETURN t
ORDER BY t.timestamp DESC
LIMIT 1;
```

```
1 MATCH (t:Transaction)
2 WHERE t.method_id IN ['deposit', 'withdraw']
3 RETURN t
4 ORDER BY t.timestamp DESC
5 LIMIT 1;
```

Graph Table RAW



Node details

Transaction

Key	Value
<id>	4:29aeedb7-79a8-4024-8c02-5f1919c40cdc:162
anonymity_set	22
gas_price	5
method_id	"deposit"
open_count	13
timestamp	1708647319
txid	"0x7051616b"

Started streaming 1 record after 1 ms and completed after 2 ms.